

APPENDIX G

Control theory

G.1. Introduction

Control theory relies on the concept of feedback. Feedback is used to:

1. Stabilize a system or improve the stability of a system. For example, a system that oscillates is considered stable (in the sense that the system remains bounded). A feedback control system can increase the damping in the system so that the oscillations persist for only a short time.
2. Reduce the sensitivity of a system to modeling errors. Open-loop control could be used if the system were perfectly known. Feedback corrects for errors in tracking, thus making the closed-loop system behave closer to the ideal.
3. Reject disturbances and attenuate noise. This is done by limiting the response of the system at frequencies where there is noise (usually high frequencies) and by tightly controlling the system (applying significant restoring control) where there are disturbances.
4. Change the transient response. Feedback can speed up or slow down the response as needed. Oscillatory behavior can be reduced.

This chapter describes control-system design. The fundamental approach used throughout is loop shaping. This is applied both for continuous and discrete-time control systems. We motivate the problem in the next section by applying control-system concepts to a very simple system.

G.2. Simple control system

Assume that you want to control the angular rate of a rotor. Further, assume that you can measure that rate with a perfect rate sensor and have a perfect torque actuator. Neither can saturate so the sensor can measure arbitrarily high rates and the torque actuator can produce arbitrarily large torques. In addition, neither sensor nor actuator has any noise and both have infinite resolution. The dynamical system is

$$\begin{aligned} I\dot{\omega} &= T_c + T_d \\ \gamma &= \omega \\ u &= \omega_c \end{aligned} \tag{G.1}$$

where γ is the measured value of rate, u is the command, T_d is the disturbance torque, and T_c is the control torque. A simple control law is

$$T_c = -k(\omega - \omega_c) \quad (\text{G.2})$$

where the forward gain k is in units of N m/rad/s. Combining the equations, we get the closed-loop dynamical equation

$$I\dot{\omega} + k\omega = k\omega_c + T_d \quad (\text{G.3})$$

We can solve this analytically to see how well the control system performs. In this case we are interested in disturbance rejection and command following. First, divide by the inertia

$$\dot{\omega} + \frac{k}{I}\omega = \frac{k}{I}\omega_c + \frac{T_d}{I} \quad (\text{G.4})$$

The unforced (homogeneous) solution is

$$\omega_h = Ae^{-\frac{k}{I}t} \quad (\text{G.5})$$

The forced (particular) solution is

$$\omega_p = \omega_c + \frac{T_d}{k} \quad (\text{G.6})$$

Given the initial condition ω_0 the total solution to a simultaneous step in command and step torque disturbance is

$$\omega = (\omega_0 - \omega_c - \frac{T_d}{k})e^{-\frac{k}{I}t} + \omega_c + \frac{T_d}{k} \quad (\text{G.7})$$

The rate will approach the commanded rate based on the damping time constant that is proportional to k . The disturbance is attenuated by the factor k . As a consequence, a large k is better. However, large k implies large torques for small errors, meaning our actuator has to produce a lot of torque and will consume a lot of energy.

This simple system can be represented as a block diagram (see Fig. G.1). The block diagram makes use of Laplace-transform notation, where $1/s$ implies integration. If we write the open-loop equations in Laplace-transform notation we get the equations

$$\omega(s) = \frac{1}{Is}(T_d(s) + k(\omega_c(s) - \omega(s))) \quad (\text{G.8})$$

$$\omega(s) \left(1 + \frac{k}{Is}\right) = \frac{T_d(s)}{Is} + \frac{k\omega_c(s)}{Is} \quad (\text{G.9})$$

$$\omega(s) = \frac{T_d(s)}{Is + k} + \frac{k\omega_c(s)}{Is + k} \quad (\text{G.10})$$

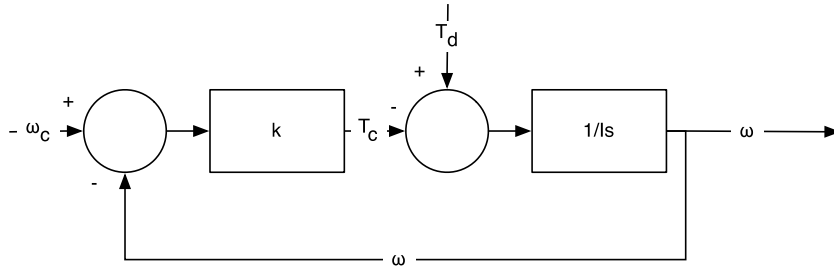


Figure G.1 Rate-control block diagram.

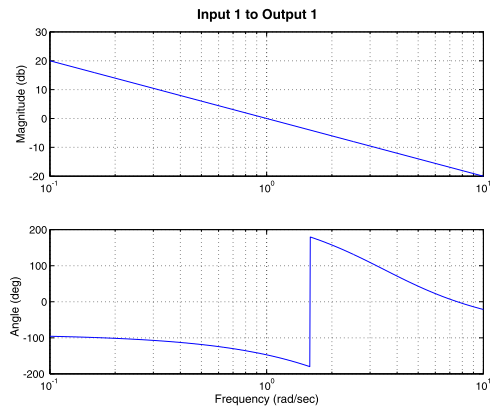
This example can be studied using Bode plots, Nichols diagrams, and root-locus plots. The state-space equations for the plant and the controller when we are looking at the disturbance input are

$$\begin{array}{cccc} a = 0 & b = 1/I & c = 1 & d = 0 \\ a_c = 0 & b_c = 0 & c_c = 0 & d_c = k \end{array} \quad (\text{G.11})$$

```

1 gain = [1 10 100];
2 w = logspace(-2,3);
3 for k = 1:3
4     [a,b,c,d] = Series(0, 1, 1, 0, [], [], [], gain
5         (k));
6     [mag(k,:),ph(k,:)] = FResp(a,b,c,d,1,1,w);
7 end
8 yL = {'Gain_(db)', 'Phase_(deg)'};
9 h = Plot2D(w, [20*log10(mag);ph], 'Frequency_(
10     rad/s)',...
11     yL, 'Bode_Plot', 'xlog', {'[1:3]',
12         [4:6]});
13 legend('gain_1=1', 'gain_10=10', 'gain_100=100')
14 set(h,'color',[1 1 1]);

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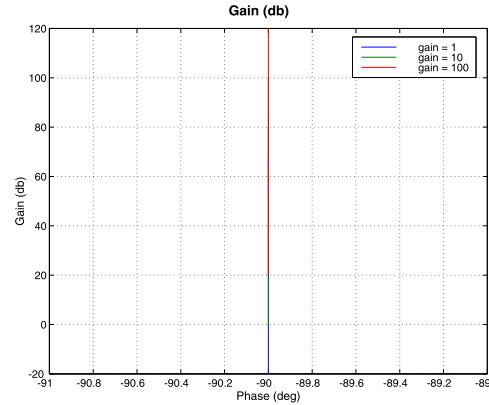
Example G.1: Bode plot.

Example G.1 shows the open-loop equations when $I = 1$ and $k = 1, 10, 100$. Bode and Nichols plots show the same information. The open-loop phase shift is constant at -90° . As the gain increases, the closed-loop gain increases. This moves the Bode plot up in the plot, thus increasing the crossover frequency, which is the frequency beyond which feedback has no influence on the closed-loop response. As the phase shift is always -90° the system is stable for all gains. In real systems this is never true because all real systems have delays and any delay will cause the system to go unstable for some value of the gain. Example G.3 shows the root-locus plot for $k = 0$ to 100 for the same system. The root-locus plot gives the closed-loop response. This plot shows that as the gain

```

1 gain = [1 10 100];
2 for k = 1:3
3     [a,b,c,d] = Series(0, 1, 1, 0, [], [], [], gain
4         (k) );
5     [mag(k,:),ph(k,:)] = FRsp(a,b,c,d);
6 end
7 h = Plot2D( ph, 20*log10(mag), 'Phase(deg)', '
8     Gain(db)');
9 legend('gain=1', 'gain=10', 'gain=100')
10 set(h,'color',[1 1 1]);

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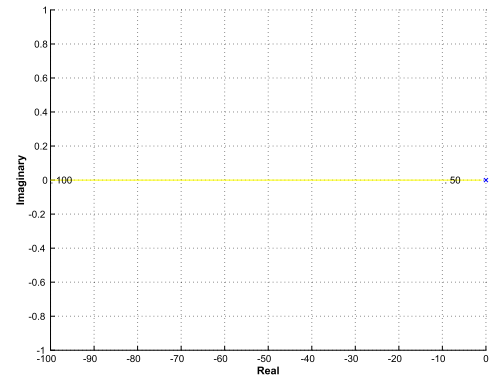


Example G.2: Nichols plot.

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1 [a,b,c,d] = Series(0, 1, 1, 0, [], [], [], 1);
2 g = statespace(a,b,c,d);
3 gain = logspace(0,2);
4 RootLocus(g, gain);
5 PrintFig(1,1,1,'RootLocus')

```



Example G.3: Root-locus plot.

goes up the closed-loop pole becomes more negative real, i.e., it becomes faster. The imaginary part remains zero. More information on single-input–single-output design is given later in this chapter.

G.3. The general control system

Fig. G.2 shows a general block diagram of a control system. r is the command input to the system, m is the measurement noise input, γ is the system-measured output, d_1 is the disturbance that is measured and used for feedforward purposes, d_2 is the unmeasured disturbance, R is the command prefiltering block, K is the control block, Q is the feedback block, F is the feedforward block, and P is the plant. P includes all plant dynamics including those of the actuators and sensors. If the system is implemented

digitally, K and F include samplers on the inputs and sample-and-holds, or pulsewidth modulators, etc. on the outputs. E is the identity matrix.

Often, unity feedback configurations are chosen in which $Q = 1$. If r is always 0 then the system is a regulator and R is zero. Many control systems do not have disturbance feedback and $F = 0$. Sometimes the disturbances inputs are put after the plant and the relevant plant dynamics are included in the disturbances.

This block diagram applies equally well to systems with multiple inputs and outputs. It is important to remember that when dealing with the multioutput system (which all real systems are), not all outputs are measured by control sensors, but may still be important to the performance.

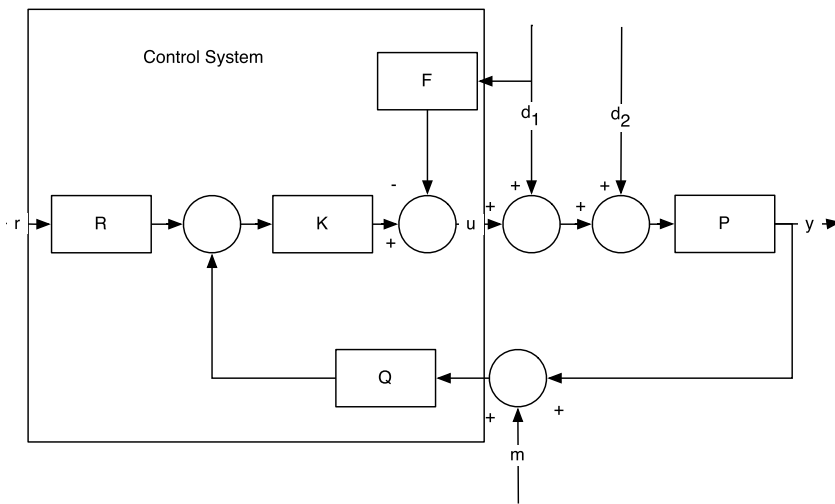


Figure G.2 General control-system block diagram.

G.4. Fundamental relationships

In this section we will derive algebraic relationships between the inputs and outputs of Fig. G.2. The error e is

$$e = Rr - Q(m + y) \quad (\text{G.12})$$

the output, y is

$$y = P((EW - F)d + u) \quad (\text{G.13})$$

and the control is

$$u = Ke \quad (\text{G.14})$$

The system inputs are m , r , and d . u and e are intermediate quantities. Solving for y in terms of the inputs

$$(E + PKQ)y = P(E - F)d + PK(Rr - Qm) \quad (\text{G.15})$$

The feedforward matrix attenuates the disturbance d and may modify its dynamics. For example, if d were a step input and F were a low-pass filter, d would then look like the product of a step and an exponential decay. Since F cannot equal E without drawing infinite power, some of d will always pass through. Therefore no generality is lost if F is dropped from further discussion since the magnitude and dynamics of d are not significant in this discussion.

Q modifies the characteristics of the feedback path. If we set

$$K = KQ^{-1} \quad (\text{G.16})$$

and

$$R = QR \quad (\text{G.17})$$

then Q disappears. Q may give some design flexibility when mechanizing the control system, but from a strictly algebraic point-of-view can be eliminated.

Eliminating Q and F and multiplying both sides by $E + PK$ gives

$$y = -(E + PK)^{-1}(Pd + PK(Rr - m)) \quad (\text{G.18})$$

where

$$F_o = E + PK \quad (\text{G.19})$$

is known as the output-return difference. This is because if the feedback loop were broken at the output of the plant, P , then F_o would be E , the identity matrix. The return difference shows the effect of feedback on the output of the plant. In Eq. (G.18) note that r and m are indistinguishable to the output, except that r is multiplied by the prefiltering matrix R .

The inverse of (G.19) is known as the sensitivity, S . It shows how sensitive the output is to the product of the plant dynamics and the disturbances. If the sensitivity is small, the output will be insensitive to disturbances. If it is one, the product of the plant dynamics and the disturbances will be passed unattenuated to the output. The sensitivity is one when the loop is opened so the sensitivity (at any given frequency) shows how close the system is to being open loop at that frequency. Eq. (G.18) can be rewritten as

$$y = Spd + G_c r - Tm \quad (\text{G.20})$$

where G_c is the transmission function and equals

$$G_c = TR \quad (\text{G.21})$$

and determines how well y will track r . Ideally, G_c would equal E , by letting

$$R = T^{-1} \quad (\text{G.22})$$

but again that would require infinite power and is not possible. T is the closed-loop transfer function

$$T = SPK \quad (\text{G.23})$$

and determines how well the noise is attenuated. Ideally it should equal zero. It is sometimes called the complementary sensitivity because

$$S + T = E \quad (\text{G.24})$$

where

$$S = (I + PK)^{-1} \quad (\text{G.25})$$

and

$$T = PK(I + PK)^{-1} \quad (\text{G.26})$$

This leads to the fundamental trade-off in control design. Ideally, S and T would be equal to zero everywhere but since their sum is the identity matrix, this is not possible. At best, and what control design is all about, S can be made small where P_d is large and T small where m is large. This works in practice since P_d is usually large at low frequencies and small at high frequencies, while sensor noise is generally high at high frequencies, and low at low frequencies except for a component near dc (known as a bias) which varies slowly with time. Table G.1 summarizes the relationship between PK , T , and S . Example G.4 shows a double-integrator plant, which represents a single axis of a vehicle in its simplest form. At $\omega = 0$ it has infinite gain and the gain rolls off as $1/\omega^2$. It is clear how important P is when evaluating the effect of disturbances. In this case, any steady-state disturbance will eventually cause the output to grow very large.

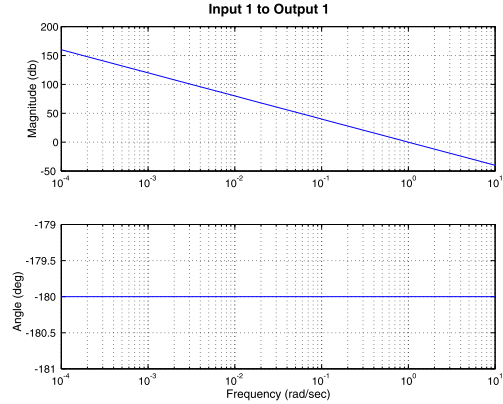
Table G.1 Relationships between PK , T , and S .

	<i>PK small</i>	<i>PK large</i>
<i>T</i>	0	<i>E</i>
<i>S</i>	<i>E</i>	0

```

1 a = [0 1; 0 0]; b = [0; 1];
2 c = [1 0]; d = 0;
3 FRsp(a,b,c,d,1,1,logspace(-4,1))

```



Example G.4: Double-integrator plant.

It is also important to keep u small, since that determines how much power is consumed, how large the actuators must be, etc. The control output is

$$u = (E + KP)^{-1} K(Rr - m - Pd) \quad (\text{G.27})$$

The factor $E + KP$ is called the input return difference (denoted by F_i) since it shows the effect of feedback on the input to the plant from the control. Note that K and P are transposed from the output return difference function. If the plant is single-input–single-output K and P are scalars so the order does not matter. For multiinput–multioutput plants, it does make a difference. If $KP \gg E$ Eq. (G.27) becomes

$$u = P^{-1}(Rr - m) - d \quad (\text{G.28})$$

This says that if the control is fast enough, it will exactly cancel the disturbances, d . The reference and the noise inputs will be amplified by the inverse of the plant. Thus in the double-integrator example, high-frequency references and noise will require more command energy than low-frequency ones.

Finally, look at the tracking error, which is defined as

$$e = r - y \quad (\text{G.29})$$

Substituting,

$$e = (E - TR)r - (E - T)Pd + Tm \quad (\text{G.30})$$

where T should equal E to eliminate tracking errors due to disturbances, but that would cause all noise to pass through. TR should equal E to eliminate tracking errors due to the reference input. Of course, for TR to equal E , R would have to be the inverse of T ,

the closed-loop transfer function, which is not physically possible. Nonetheless, these goals can be achieved over different parts of the frequency spectrum. Thus at frequencies where the reference is likely to have energy TR should equal E . At disturbance frequencies, T should equal E , and everywhere else T should equal zero to minimize noise, see Table G.2.

Table G.2 What quantities should be small to compensate for large inputs. The quantities in the table make the quantities on the left small.

	m large	P_d large	r large
γ	T	S	N/A
e	T	$E - T$	$E - TR$
u	$F_i K$	$F_i K$	$F_i K R$

G.5. Tracking errors

In Eq. (G.30) if $R = E$ then the transfer function between the reference input and the error is just the sensitivity S . Rr is just a modification of the input-signal dynamics, therefore the sensitivity will determine the tracking error to the output of r .

Disturbances and reference signals can be characterized in the frequency domain just as plants and controllers can. Typical inputs are given with their transfer functions in Table G.3. R can be used to modify the dynamics of the inputs. This can make it

Table G.3 Inputs.

Input	Time Domain	Frequency Domain
Impulse	d	1
Step	A	A/s
Ramp	At	A/s^2
Sinusoid	$A \sin \omega t$	$A\omega/(s^2 + \omega^2)$

easier for the closed-loop system to follow its input.

For example, given the simple single-input-single-output plant with a gain of k , a $1/s$ plant, an input transfer function of R and a ramp input, the error is

$$\frac{e}{A} = \frac{R}{s^2} \left(\frac{s}{s+k} \right) \quad (\text{G.31})$$

where the quantity in parentheses is the sensitivity. The steady-state tracking error can be found using the final-value theorem, which states that

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s f(s) \quad (\text{G.32})$$

R must be of order s for the tracking error to go to zero in steady state. A suitable transfer function would be

$$R = \frac{s}{\tau s + 1} \quad (\text{G.33})$$

An alternative is to make the controller a proportional-integral controller.

$$K = k \left(1 + \frac{1}{\tau s} \right) \quad (\text{G.34})$$

Then, the sensitivity would become

$$S = \frac{\tau s^2}{\tau s^2 + k\tau s + 1} \quad (\text{G.35})$$

which has the same desired effect. In general, for the tracking error to approach zero asymptotically, the sensitivity must be of the same order as the input.

G.6. State-space closed-loop equations

Fig. G.3 shows the block diagram in the unity feedback configuration without the command prefiltering.

The control block and the plant can both be represented by sets of first-order differential equations

$$\dot{x}_p = A_p x_p + B_p u_p \quad (\text{G.36})$$

$$y_p = C_p x_p + D_p u_p \quad (\text{G.37})$$

$$\dot{x}_c = A_c x_c + B_c u_c \quad (\text{G.38})$$

$$y_c = C_c x_c + D_c u_c \quad (\text{G.39})$$

$$u_c = r - m - y_p \quad (\text{G.40})$$

$$u_p = d + K y_c \quad (\text{G.41})$$

The state equations for the closed-loop plant are

$$\dot{x} = Ax + B_d d + B_m(r - m) \quad (\text{G.42})$$

and the measurement equation is

$$y = Cx + D_d d + D_m(r - m) \quad (\text{G.43})$$

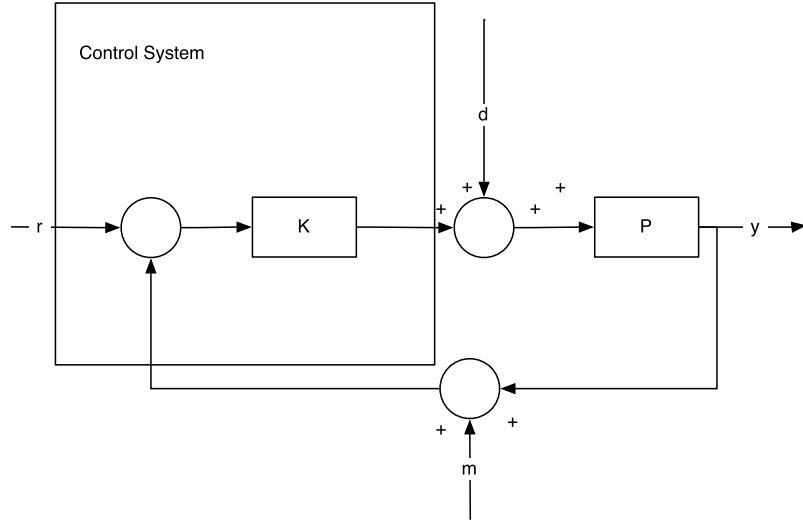


Figure G.3 Control-system block diagram.

where

$$x = \begin{bmatrix} x_p \\ x_c \end{bmatrix} \quad (\text{G.44})$$

$$A = \begin{bmatrix} A_p - B_p K D_c F C_p & B_p K (C_c - D_c F D_p K D_c) \\ -B_c F C_p & A_c - B_c F D_p K C_c \end{bmatrix} \quad (\text{G.45})$$

$$B_d = \begin{bmatrix} B_p (E - K D_c F D_p) \\ -B_c F D_p \end{bmatrix} \quad (\text{G.46})$$

$$B_m = \begin{bmatrix} B_p K D_c (F D_p K D_c - E) \\ -B_c (E - B_p K D_c F D_p) \end{bmatrix} \quad (\text{G.47})$$

$$C = \begin{bmatrix} F C_p \\ F D_p K C_c \end{bmatrix} \quad (\text{G.48})$$

$$D_d = F D_p \quad (\text{G.49})$$

$$D_m = -F D_p K D_c \quad (\text{G.50})$$

$$F = (E + D_p K D_c)^{-1} \quad (\text{G.51})$$

Adding the prefilter plant is straightforward. r is replaced by γ_r and an additional set of state equations

$$\dot{x}_r = A_r x_r + B_r r \quad (\text{G.52})$$

$$\gamma_r = C_r x_r + D_r r \quad (\text{G.53})$$

$B_m C_r$ is added to the state equations for x_p and x_c . $B_m D_r$ becomes a new input matrix with the third block row equal to B_r . γ is unchanged. The final state equation becomes

$$\dot{x} = \begin{bmatrix} A_{pp} & A_{pc} & B_{mp}A_r \\ A_{cp} & A_{cc} & B_{mc}A_r \\ 0 & 0 & A_r \end{bmatrix} \begin{bmatrix} x_p \\ x_c \\ x_r \end{bmatrix} + \begin{bmatrix} B_{pm} & B_{pd} & B_{mp}D_r \\ B_{cm} & B_{cd} & B_{mc}D_r \\ 0 & 0 & D_r \end{bmatrix} \begin{bmatrix} m \\ d \\ r \end{bmatrix} \quad (\text{G.54})$$

G.7. Approaches to robust control

G.7.1 Introduction

This section outlines several approaches to robust control. Robustness is the property of a closed-loop system by which it remains stable and delivers acceptable performance in the face of unmodeled perturbations to the plant, disturbances, and noise. These perturbations may be as simple as unmodeled gain variations in an actuator, or as complex as variations in the structure of the plant model. The two forms of robustness are known as stability and performance robustness. When discussed in the context of multivariable control systems, an additional important property, sometimes known as the integrity of the system, is its stability in the face of loop failures.

Since the field of robust control is very active, it is not possible to describe all methodologies. Noticeably absent in this section are detailed discussions of adaptive control and variable-structure control. Adaptive control has been used with a fair degree of success in the process-control industry, but does not, at present, appear suitable for fully autonomous control systems. Variable-structure control has been proposed for control systems and has been successfully applied to induction motor controllers, among other applications, but since it is a nonlinear control approach it requires extensive simulations to validate, making it costly to implement and impractical given the schedule constraints on most flight software.

G.7.2 Modeling uncertainty

Robust control techniques require models of uncertain dynamics. The form of the model will directly determine the conservativeness of the resulting control system. The more conservative a design is, the poorer its performance will be with the nominal dynamics; this can lead to a need for better, and more expensive, actuators and sensors.

In classical control design, uncertainty is often represented by loop-gain uncertainty at the phase-crossover frequencies and phase uncertainty at the gain-crossover frequencies. These translate into the gain and phase margins. Complex systems, such as those with bending or nutation modes near or within the control-system bandwidth, have multiple crossovers. For example, a designer of a single-input-single-output (SISO) system might decide that the frequency of the first bending-mode frequency was so uncertain that the loop-crossover frequency had to be significantly below the nominal

first bending-mode frequency so that the bending mode was gain stabilized. Gain stabilization implies that the mode peak is below 0 dB and no amount of phase shift can cause it to destabilize the closed-loop system. Naturally, this leads to a lower performance since the bandwidth is restricted and lower than the frequency of the bending mode. If the modal frequency were better known, a notch filter might be employed or the mode might be phase stabilized. These solutions would lead to better performance.

A standard representation of uncertainty is illustrated in Fig. G.4. All uncertainty in

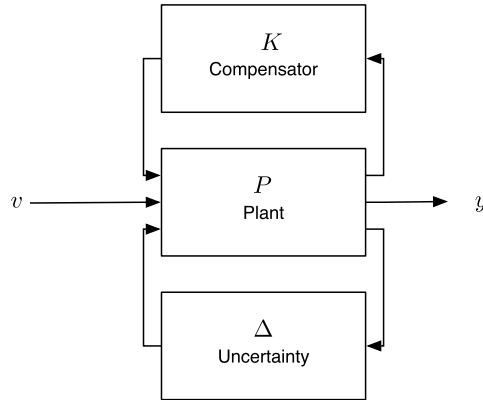


Figure G.4 Standard representation of uncertainty.

the plant is pulled out into the transfer matrix Δ . The simplest approach is to permit Δ to be unstructured and to only restrict its size that is measured by the matrix norm

$$\|\Delta(s)\|_{\infty} \quad (\text{G.55})$$

This expression is the maximum value of the norm of $\Delta(s)$. For a single-input–single-output transfer function, it would be the peak magnitude on a Bode plot.

The size can also be made frequency dependent by pre- and postmultiplication by frequency-dependent weighting functions. The uncertainty may be treated as an additive perturbation to the plant or a multiplicative uncertainty. In the latter case, which is often more useful since the size of Δ can be specified as a percentage variation in the plant dynamics, the uncertainty must be specified as either an input or output multiplicative uncertainty since matrix multiplication does not commute.

When a robust design is based on the size of Δ , the system will remain stable regardless of the variations within Δ as long as the size remains within the prespecified limits. This can lead to a conservative design. For example, even though it may be known that Δ is diagonal, a given size of Δ can also accommodate off-diagonal terms. Therefore a design based on the size will tend to be conservative.

A less-conservative approach is to account for the structure of Δ . Structured uncertainty methods take advantage of knowing which elements of Δ are nonzero. Δ takes on a block structure with submatrices representing unstructured uncertainty. The uncertainty model might be

$$\Delta = \begin{bmatrix} \delta_r & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \Delta_3 \end{bmatrix} \quad (\text{G.56})$$

where Δ_3 represents the plant uncertainty and δ_r is the roll-channel gain uncertainty. The pitch-channel uncertainty is zero. This model, while better than the unstructured model, is still conservative because the elements of Δ may be known to be purely real, yet any general bound on Δ will permit complex elements. As a consequence, a design employing this form of structured uncertainty will accommodate phase uncertainty that may not be physically possible.

A more accurate representation of uncertainty is to compute the set of possible frequency responses of the plant at each frequency. For example, for a double integrator with a flexible mode and collocated sensor and actuator, the plant is

$$\frac{\Theta(s)}{T(s)} = \frac{s^2 + 2\zeta\omega s + \omega^2}{Is^2(s^2 + \gamma(2\zeta\omega s + \omega^2))} \quad (\text{G.57})$$

where γ is the modal coupling factor

$$\gamma = 1 + \frac{I_\mu}{I} \quad (\text{G.58})$$

where I_μ is the modal inertia and γ is ≥ 1 . The uncertainty template would be found by computing all values of the transfer function for the specified range of the parameters. This permits a direct representation of the uncertainty with all of the gain and phase information. Generally, I and ω are known accurately, but γ and ζ are not. Nonetheless, the transfer function varies in a deterministic way with these parameters and it is easy to find a set of uncertain plants for realistic ranges of γ and ζ . There is nothing to prevent the use of a nonlinear relationship between the gain, phase, and plant parameters.

The second form of robustness, performance robustness, can be related to the structured uncertainty described above. Given the model in Fig. G.5 the transfer function is

$$\begin{bmatrix} \gamma \\ x \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} \begin{bmatrix} v \\ z \end{bmatrix} \quad (\text{G.59})$$

and performance robustness can be defined as

$$\|Q_{11} + Q_{12}\Delta(I - Q_{22}\Delta)^{-1}Q_{21}\|_\infty < 1 \quad (\text{G.60})$$

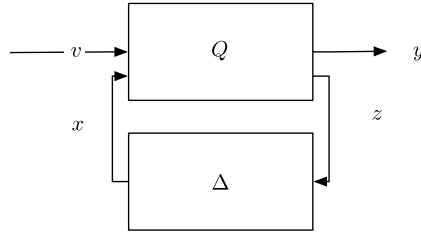


Figure G.5 Compensated plant.

$$\|Q_{22}\|_{\mu} < 1 \quad (\text{G.61})$$

$$\Delta = \text{diag}(\Delta_1, \dots, \Delta_n) \quad (\text{G.62})$$

$$\|\Delta_i\| \leq 1 \quad (\text{G.63})$$

where

$$\|Q_{22}\|_{\mu} = \sup \mu(Q_{22}(j\omega)) \quad (\text{G.64})$$

and $\mu(Q)$ is the structured singular value of Q for a block diagonal Δ . Structured uncertainty can be used both to design for performance robustness and to make designs less conservative when designing for stability robustness.

These uncertainty models do not accommodate loop failures and the resulting designs may be unstable when loops or combinations of loops are broken. For a single-input–single-output system, the open-loop performance and stability are just that of the open-loop plant. For a multiinput–multioutput design, the performance will be that of the remainder of the closed-loop plant. With three loops in a 3-axis controlled vehicle, there are three possible systems with one loop broken, and six if two loops break.

G.7.3 Control-structure design

The pairing of inputs and outputs in many systems is of critical importance in designing a low-cost robust system with good performance. Since controls often affect many modes of the system effective pairing can reduce controller complexity.

G.7.4 Nyquist-like techniques

Nyquist-like techniques are extensions of single-input–single-output design for multivariable systems. Three common techniques are sequential loop closing, the characteristic-loci method, and the Nyquist-array approach. These approaches all replace the multivariable problem with a sequence of SISO problems. In each method, the first stage is to decouple the system across a range of frequencies with either a pure gain or dynamic compensator. If the latter is used, there is always the danger of canceling lightly damped poles that often exist in vehicle-control problems. A simple example of

a decoupling stage is in the case of a vehicle with significant off-diagonal inertia terms. The state equations for the system are

$$\dot{x} = \begin{bmatrix} 0 & E \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ I^{-1} \end{bmatrix} u \quad (\text{G.65})$$

where E is a 3×3 identity matrix. If the feedback structure is chosen to be of the form

$$u = IKx \quad (\text{G.66})$$

the state equations become

$$\dot{x} = \begin{bmatrix} 0 & E \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ E \end{bmatrix} Kx \quad (\text{G.67})$$

and K can be designed for the three decoupled loops.

In sequential loop closing, the fastest loops (such as a reaction-wheel tachometer loop) are closed first. If the separation of loop bandwidths is sufficient, this can lead in a straightforward fashion to the final design. When this is not the case, the design usually proceeds in an ad-hoc manner and design decisions made when closing the first loop may have deleterious effects later. The only means available for reducing interaction between the loops is to apply high loop gains.

A second method is a characteristic-locus method that treats the loci of a transfer-function matrix much as one would treat the locus of a single-input-single-output system. With this method, a single-input-single-output controller is designed for each locus.

In the Nyquist-array method, the first step is to introduce a compensator in series with the plant that makes the product of the two diagonally dominant over some range of frequencies. When this is accomplished, the system will behave like a set of SISO loops and the design can proceed.

G.7.5 LQG methods

Linear quadratic Gaussian methods (LQG) are a type of state-space method. The form of an LQG controller is

$$u = -K\hat{x} \quad (\text{G.68})$$

and the estimated state is obtained from

$$\dot{\hat{x}} = (A - BK)\hat{x} + Bu + L(\gamma - C\hat{x}) \quad (\text{G.69})$$

given the state-space model

$$\dot{x} = Ax + Bu + \Gamma w \quad (\text{G.70})$$

$$y = Cx + v \quad (\text{G.71})$$

where w and v are Gaussian white-noise sources for which

$$E(w w^T) = W \quad (\text{G.72})$$

$$E(v v^T) = V \quad (\text{G.73})$$

$$E(w v^T) = N \quad (\text{G.74})$$

The plant matrix A may be of the dynamic system alone, or may include additional dynamics, such as disturbance models, integrators, filters, etc. to add frequency dependence to Eq. (G.68). The gain matrix is one that minimizes the function

$$J = \lim_{T \rightarrow \infty} E \left(\int_0^T (z^T Q z + u^T R u) dt \right) \quad (\text{G.75})$$

where

$$z = Mx \quad (\text{G.76})$$

Q and R can also be functions of frequency.

An LQG controller can exhibit arbitrarily poor robustness properties. Loop-transfer recovery (LTR) was developed to recover the robustness properties of the full-state feedback controller (i.e., one with all states measured). The LTR procedure places some of the estimator eigenvalues at the zeros of the plant, and the remainder is allowed to be arbitrarily fast. This only works if the plant does not have any right-hand-plane zeros near the desired bandwidth and if the plant is square, that is the number of inputs and outputs are equal. If it has unequal numbers of inputs and outputs, LTR can still be applied by squaring the plant. For example, if there are more inputs than outputs, linear combinations of outputs can be added to make the number of outputs equal to the number of inputs. If this is done, recovery can only be applied at the output.

The LQR methodology does not address uncertainty in the plant model directly. Since uncertainty models are not part of the design procedure, it is necessary to test the system over the range of parameters expected in the plant, including higher-frequency dynamics that were neglected in the design. In addition, choosing the weighting matrices Q and R to get the desired performance is not easy, although methodologies for doing so have been suggested, making LQR labor-intensive and expensive to implement.

G.7.6 H_∞ and μ synthesis

The H_∞ (pronounced H-infinity where the H stands for Hardy space) optimal controller is defined as follows. The standard plant representation is shown below. If P is

partitioned into four parts such that

$$\begin{bmatrix} z \\ y \end{bmatrix} = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} \quad (\text{G.77})$$

the input/output relationship is (see also Fig. G.6)

$$z = [P_{11} + P_{12}K(I - P_{22})^{-1}P_{21}]w \quad (\text{G.78})$$

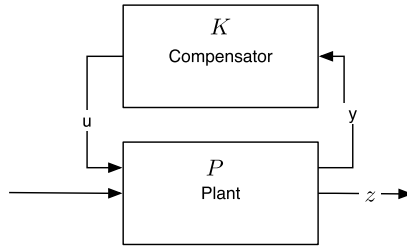


Figure G.6 Standard plant model.

Define

$$F_1(P, K) = [P_{11} + P_{12}K(I - P_{22})^{-1}P_{21}] \quad (\text{G.79})$$

The H_∞ problem is to minimize

$$\|F_1(P, k)\|_\infty \quad (\text{G.80})$$

over all realizable controllers $K(s)$ that stabilize the closed-loop system. A consistent term for the LQG problem is the H_2 problem since it involves minimizing

$$\|F_1(P, k)\|_2 \quad (\text{G.81})$$

Many control-design problems such as sensitivity minimization and robustness to perturbations can be formulated as H_∞ problems. However, performance robustness in the face of unmodeled perturbations cannot be formulated as an H_∞ problem. As noted in the above section, the performance robustness is related to the structured singular value of Q . This relationship is used as a basis for the μ -synthesis approach to robust compensator design. The structured singular value cannot be computed directly, and it is necessary to resort to numerical algorithms.

G.8. Single-input–single-output control design

G.8.1 Introduction

The fundamental approach used throughout this chapter is the frequency-domain approach to loop shaping. This is applied both for continuous controllers and discrete-time control systems. All techniques are applied to single-input–single-output control systems in this chapter.

G.8.2 Elementary loop compensation

G.8.2.1 First-order compensators

Control compensators can be created in a number of ways. In this section we will use first-order building blocks to build control systems. The first block of interest is the first-order transfer function

$$K(s) = K \frac{\tau_1 s + 1}{\tau_2 s + 1} \quad (\text{G.82})$$

This is sometimes known as a lead or lag compensator. At low frequencies, it has a gain of K and at high frequencies, its gain is constant and is

$$K(s) = K \frac{\tau_1}{\tau_2} \quad (\text{G.83})$$

It is useful to convert (G.82) into its equivalent phase and magnitude as a function of frequency, where $s = j\omega$. The first step is to make the denominator purely real by multiplying the numerator and the denominator by the complex conjugate of the denominator.

$$K \left(\frac{\tau_1 s + 1}{\tau_2 s + 1} \right) \left(\frac{1 - \tau_1 s}{1 - \tau_2 s} \right) = K \frac{1 + \tau_1 \tau_2 \omega^2 + j(\tau_1 - \tau_2)\omega}{1 + (\tau_2 \omega)^2} \quad (\text{G.84})$$

The phase as a function of frequency is

$$\theta = \frac{(\tau_1 - \tau_2)\omega}{1 + \tau_1 \tau_2 \omega^2} \quad (\text{G.85})$$

with a peak value of

$$\theta_{peak} = \tan^{-1} \frac{\tau_1 - \tau_2}{2\sqrt{\tau_1 \tau_2}} \quad (\text{G.86})$$

at

$$\omega_{peak} = \frac{1}{2\sqrt{\tau_1 \tau_2}} \quad (\text{G.87})$$

This compensation element is most useful to add or subtract phase at a desired frequency. Therefore we want to know what the time constants should be. Solving (G.86) and

(G.87) for the time constants gives

$$\tau_2 = \sqrt{\frac{1 + \frac{1}{4} \tan^2 \theta_{peak}}{\omega_{peak}^2}} + \frac{\tan \theta_{peak}}{2\omega_{peak}} \quad (\text{G.88})$$

and

$$\tau_1 = \tau_2 + \frac{\tan \theta_{peak}}{\omega_{peak}} \quad (\text{G.89})$$

If the peak phase is negative, the high-frequency gain will be lower than the low-frequency gain. This configuration is often called a low-pass filter or lag compensator. If the peak phase is positive and the high-frequency gain is higher than the low-frequency gain, this is known as a high-pass filter or lead compensator.

G.8.2.2 Generalized integrator

A generalized integrator is

$$K(s) = \frac{\tau s + 1}{s + \sigma} \quad (\text{G.90})$$

If σ and τ are zero this becomes a pure integrator. If only σ is zero, it is an integrator with high-frequency phase compensation. If only τ is zero, it is a low-pass filter with a low-frequency gain of $1/s$.

σ is nonzero when infinite gain at $\omega = 0$ is not necessary. Given a specification that the DC gain equals k , and the phase equal θ_0 (between -90° and 0°) at frequency ω , then the constants are

$$\tau = \frac{\frac{\sigma \tan \theta_0}{\omega} + 1}{\sigma - \omega \tan \theta_0} \quad (\text{G.91})$$

$$\sigma = \frac{1}{k} \quad (\text{G.92})$$

G.8.2.3 Gain compensation of a double-integrator plant

The simplest model of a vehicle is the double integrator that represents one axis of a rigid body,

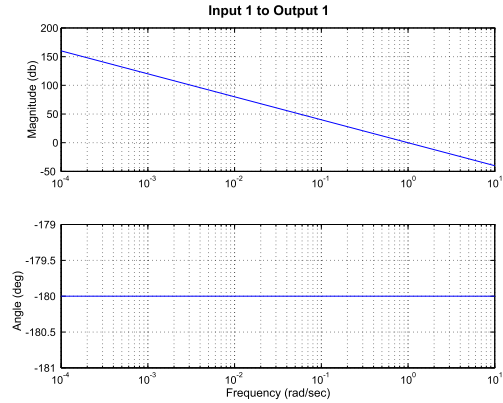
$$I\ddot{\theta} = T \quad (\text{G.93})$$

The output is the attitude and the input is torque. The Laplace transform of Eq. (G.93) is

$$\frac{\Theta(s)}{T(s)} = \frac{1}{Is^2} \quad (\text{G.94})$$

The double-integrator plant is only truly valid if the vehicle is a rigid sphere. Nonetheless, the principles used to compensate it are valid for more complex systems. First, look at the Bode plot of the plant, with an inertia of 1, in Example G.5. The gain is infinite

```
1 a = [0 1; 0 0]; b = [0;1];
2 c = [1 0]; d = 0;
3 Fresp(a,b,c,d,1,1,logspace(-4,1))
4 PrintFig(1,1,1,'DoubleIntegrator')
```



Example G.5: Double-integrator model.

at $s = 0$. Therefore any steady input will cause the angle to go to infinity eventually. The response to inputs at other frequencies drops as ω^2 . The phase is -180 deg at the gain crossover of $\omega = 1$ rad/s. As a consequence, if we were to close the loop with negative feedback, the plant, if perturbed, would oscillate with a frequency equal to the crossover frequency. Since the phase is -180° everywhere, changing the feedback gain would just shift the frequency of the oscillation. This is clear from adding a $-Kq$ term to the right-hand side of Eq. (G.93)

$$\ddot{\theta} + \frac{K}{I} = \frac{T}{I} \quad (\text{G.95})$$

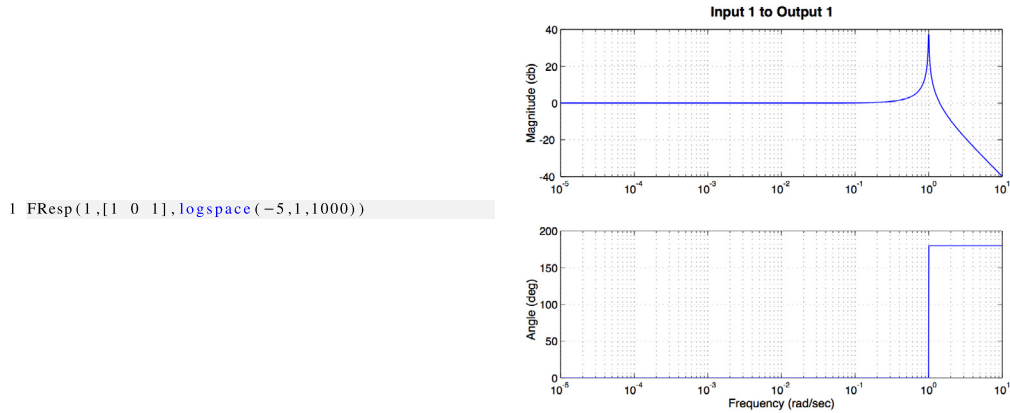
which is the equation for an oscillator with an oscillation frequency of

$$\omega = \sqrt{\frac{K}{I}} \quad (\text{G.96})$$

The first question is why bother using feedback at all? Assume that the design requirements state that the only input is a 0.01-N m torque at 0.1 rad/s and that the peak attitude error must be less than 0.001 rad. If the inertia is greater than 1000 kg m², the open-loop plant meets specifications and no further work is necessary. If the inertia were too small, we could meet the specification by adding inertia.

The primary reason for using feedback is to deal with uncertainty, or errors, in the models of the plant, the disturbances, and the specifications. The latter should not be easily dismissed. It is not unusual for specifications to change after the design is

done. In this simple example, that might mean a tighter pointing requirement after the vehicle had been built, and increasing its inertia is no longer a viable option. To see how feedback reduces sensitivity to uncertainty, add unity feedback ($K = 1$) to the above vehicle and look at the magnitude plot of the closed-loop system. From $\omega = 0$ to $\omega = 0.2$



Example G.6: Torque transmission.

the angle is the same for all frequency inputs. Thus for disturbances at frequencies lower than the resonant frequency, we need not be concerned about their frequency! This is in sharp contrast to the open-loop case where the angle went up as the frequency went down. Thus if the specification were that the response to any single disturbance between 0.001 and 0.4 rad/s was to be less than 0.01 rad, it would be much easier to meet the specification with feedback. At frequencies greater than 2 rad/s, the transfer function between angle and torque is nearly the same as in the open-loop case. This can be seen from the closed-loop transfer function

$$\frac{\Theta(s)}{T(s)} = \frac{1}{Is^2 + K} \quad (\text{G.97})$$

where the first term in the denominator dominates at large frequencies. The range of frequencies for which the control is effective, in this case from 0 to 2 rad/s, is called the bandwidth of the control system. The system is now also much less sensitive to variations in inertia. For all frequencies for which $K \gg Is^2$, the transfer function is

$$\frac{\Theta(s)}{T(s)} = \frac{1}{K} \quad (\text{G.98})$$

Thus the angle response to a disturbance torque is independent of the inertia and depends only on K , the feedback gain. Unfortunately, in this design we have greatly increased our sensitivity to disturbances (or noise) near 1 rad/s due to our control system. Table G.4 summarizes the performance of the closed-loop system.

Table G.4 Control-system performance summary.

Frequency Range	Torque Transmission	Characteristic
$0.01 \leq \omega \leq 0.2$	$\frac{\theta(\omega)}{T(\omega)} = \frac{1}{K}$	Closed loop
$0.02 \leq \omega \leq 0.2$	$\frac{\theta(\omega)}{T(\omega)} = \left \frac{1}{K - I\omega^2} \right $	Closed loop
$\omega \geq 2$	$\frac{\theta(\omega)}{T(\omega)} = \frac{1}{I\omega^2}$	Open Loop

In summary, this control system has minimal effect on frequencies greater than the bandwidth, which causes the torque to angle transmission to be dominated by the feedback gain at frequencies lower than a decade below the bandwidth, and causes the response to be very sensitive to disturbances or noise at frequencies near the resonant frequency of the system.

In this system, we applied a gain K as the feedback element. This implies that any input, no matter how high its frequency, is amplified by a factor K . In practice, it is not possible to design a controller with a flat response to infinity, but the bandwidth of the sensors and actuators usually exceeds that of the control system, which is usually sufficient.

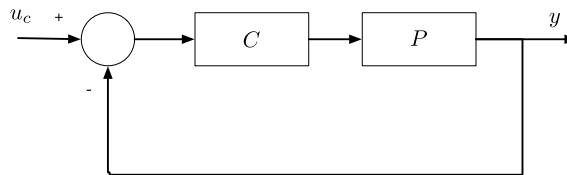
G.8.2.4 More complex compensation of a double-integrator plant

Table G.5 gives a more interesting specification for our control system in terms of both noise and disturbance.

Table G.5 Control-system specification.

Specification	Frequency Range	Input at any one Frequency	Maximum Angle	Gain
Noise	$\omega \geq 1.0$	0.01 rad	0.0001 rad	-40 db
Disturbance	$\omega = 0$	0.0001 N m	0.0000 rad	∞
Disturbance	$0 < \omega \leq 0.001$	0.0001 N m	0.0001 rad	+120 db

Fig. G.7 gives the block diagram for this system. This is a unity-feedback configuration without any disturbance feedforward and no prefiltering.

**Figure G.7** Unity-feedback block diagram.

The reference is assumed to be 0, and we are only interested in the control system as a regulator. The relationship between the output and the noise and disturbance inputs is

$$\theta = \frac{P}{1 + PK}d - \frac{PK}{1 + PK}m \quad (\text{G.99})$$

where PK is large, the equation becomes

$$\theta = \frac{1}{K}d - m \quad (\text{G.100})$$

Thus noise is passed into the output with a gain of 1 and the disturbance is attenuated by a factor of K . Where PK is small it becomes

$$\theta = Pd \quad (\text{G.101})$$

which is the open-loop equation.

Eq. (G.100) is a fundamental problem with feedback control. If one applies feedback to attenuate disturbances at a given frequency, any noise at that frequency will be passed into the output.

It is also worth noting that the product of PK is important, not just K alone. Thus the roll-off at high frequencies exhibited by all physical systems means that at high frequencies, PK will go to zero.

The gain specifications are found by substituting the values for the ratios of the output to either the disturbance or noise and computing the product PK . In the case of the disturbance, we must also know P at the frequency of interest. The specifications require -100 dB of roll-off in three decades. For stability purposes, the roll-off must be less than $1/s^2$, or 40 dB/decade at the crossover since for a minimum-phase system a $1/s^2$ slope; this means the phase shift is -180° .

The design of this control system starts by adding an integrator to meet the steady-state offset requirement. This gives the system a gain slope of -60 dB/decade at low frequencies. This must transition to greater than -40 dB/decade at the gain crossover. The form of the integrator is

$$\frac{s + 0.1}{s} \quad (\text{G.102})$$

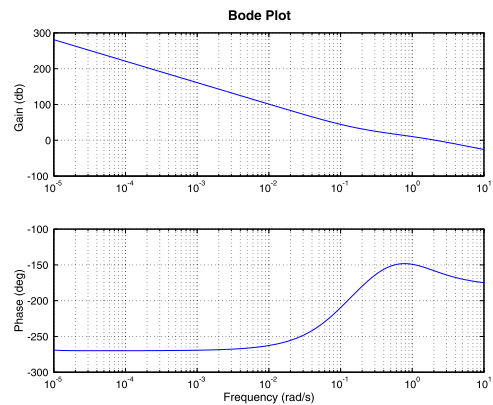
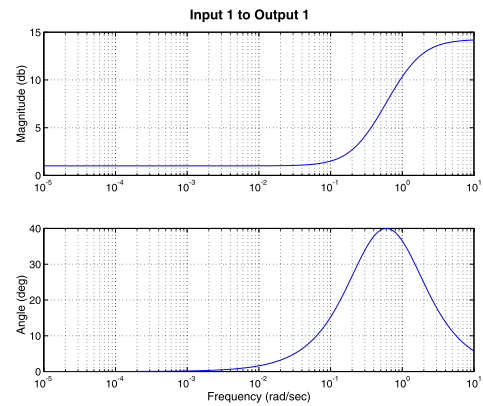
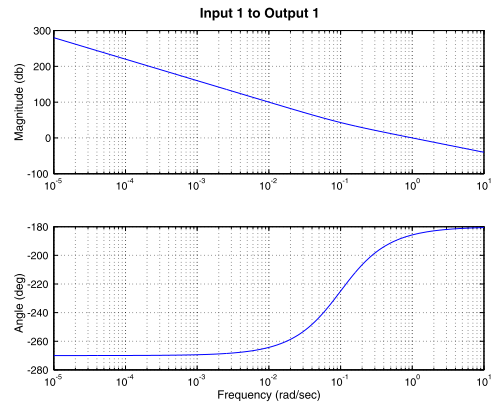
The numerator term removes the phase lag of the integrator at frequencies higher than 0.1 rad/s, thus reducing its impact at the crossover frequency.

At 0.01 rad/s, the integrator only contributes -10° of phase shift to the system. The system remains unstable because, at the crossover (to the right of the graph), the phase will be slightly lower than -180° due to the pole at the origin introduced by the integrator. Our specification calls for a gain of 110 dB at 0.001 rad/s and a gain of -40 dB at 1 rad/s. The difference between those gains is 150 dB. The gain change is correct. However, the plant is unstable.

```

1 % Plant
2 a = [0 1; 0 0]; b = [0; 1];
3 c = [1 0]; d = 0;
4 w = logspace(-5, 1, 1000);
5
6 % Integrator
7 [aI, bI, cI, dI] = ND2SS([1 0.1], [1 0]);
8 [a, b, c, d] = Series(a, b, c, d, aI, bI, cI, dI);
9 FResp(a, b, c, d, 1, 1, w)
10
11
12 % Lead
13 [aL, bL, cL, dL] = LeadLag(0.6, 40, 1);
14 FResp(aL, bL, cL, dL, 1, 1, w)
15
16
17 % Final system
18 [a, b, c, d] = Series(a, b, c, d, aL, bL, cL, dL);
19 [m, p] = FResp(a, b, c, d, 1, 1, w, 'unwrap');
20 k = find(p < -270); p(k) = p(k) + 360;
21 yL = {'Gain_ (db)', 'Phase_ (deg)'};
22 h = Plot2D(w, [20*log10(m); p], ...
23           'Frequency_ (rad/s)', ...
24           yL, 'Bode_Plot', 'xlog');

```



Example G.7: Loop-compensation example.

To correct this, we would like to adjust the phase at the crossover without changing the gain much. This can be done with lead compensation. We need to add about $+60^\circ$ of phase shift at 0.4 rad/s.

The use of standard control elements, leads, lags, and PI can be used to achieve the desired performance. This procedure requires iteration on the design. After loop compensation, the control loop needs to be tested to make certain that sufficient control authority is available for the system.

G.9. Digital control

G.9.1 Introduction

Vehicle control systems are usually implemented on digital computers. This section discusses digital control-system design techniques.

G.9.2 Modified continuous design

G.9.2.1 Introduction

Most control engineers are first taught the control of continuous systems in which the plant to be controlled is described by differential equations

$$\dot{x} = f(x, u, t) \quad (\text{G.103})$$

and the measurements are described by the nonlinear relationship

$$y = h(x, u, t) \quad (\text{G.104})$$

The linearized version of these equations are

$$\dot{x} = Ax + Bu \quad (\text{G.105})$$

and

$$y = Cx + Du \quad (\text{G.106})$$

Corresponding to these equations is the frequency-domain representation of these equations

$$y(s) = (C(sE - A)^{-1}B + D)u(s) \quad (\text{G.107})$$

Fig. G.8 shows a block diagram of a control system and plant. The only discrete part of the system is the computer and its software, the remainder of the system is continuous. Measurements are periodically entered into the computer using a sampling mechanism and numbers representing the desired actuation are converted to continuous operations using the discrete-to-continuous conversion block.

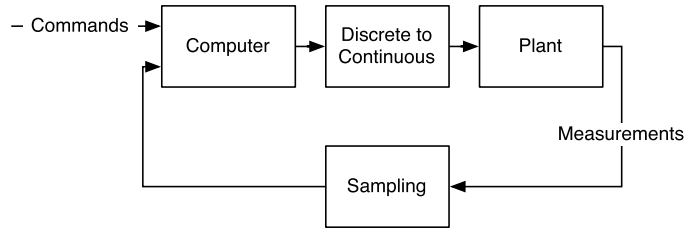


Figure G.8 Computer-controlled system.

The discrete-to-continuous conversion block may take several forms. The most common is the sample and hold. During every control period, an external register is set to the number representing the desired actuation. This value is maintained for the entire control period. For example, a register that is an input to a digital-to-analog converter that determines the input voltage to a reaction wheel might be set to the desired value. A single bit that drives a thruster-valve relay might be set and held high for the duration of the command period. Thrusters are examples of actuators that usually have only two states; these are sometimes driven by pulsewidth modulation. The output of the control system, which is a torque level for that output, is converted into a pulsewidth. If the desired torque level is u , and the control period is T , then the pulsewidth is

$$\tau = \frac{uT}{u_c} \quad (\text{G.108})$$

where u_c is the torque available from the actuator. Thus the average torque over the control period equals u . The pulsewidth will be limited by the resolution of the timing mechanism, and the speed of response of the actuator.

Since the output sampling rate does not need to be fixed, it is also possible to vary the actuation period on the output. This is known as pulsewidth–pulse–frequency modulation.

The modified frequency continuous design approach represents the conversion of the measurements to digital form, and the conversion of the commands to continuous from a continuous function. This permits the use of frequency-domain design techniques and the nearly direct conversion of the resulting algorithms into discrete form, with minimal effort. Since many modern control-design techniques are continuous-time techniques, this permits a wider variety of design approaches.

In this section, we will look at the key elements that give a continuous-time system the characteristics of a discrete-time system. These are the sampler, the delay, and the discrete-to-continuous conversion function. For the latter, we will look at the sample and hold and pulsewidth modulation.

G.9.2.2 The sampler

The sampler is represented in the continuous-time domain as a gain of $1/T$, where T is the sample period.

G.9.2.3 The delay

All digital control systems have true delays, as opposed to lags. A delay means that a signal, regardless of frequency, is always the delay time late. The Laplace transform of a delay is

$$e^{-s\Delta} \quad (\text{G.109})$$

where Δ is the delay in seconds and s is $j\omega$. This is a nonminimum-phase function. Nonminimum phase means that the phase angle is greater than the minimum possible for a function with the same magnitude. Eq. (G.109) can be expanded as

$$e^{-s\Delta} = \cos \omega\Delta - j \sin \omega\Delta \quad (\text{G.110})$$

The magnitude of this function is 1 and the phase is $-\omega\Delta$. Thus the phase decreases linearly with frequency. This is also known as an all-pass function. The minimum phase for a constant magnitude function is 0. Therefore this function is a nonminimum phase. All-pass functions are always nonminimum phase.

Any uncertainty in a delay can wreak havoc on control approaches that require precise phase knowledge at high frequencies since the phase error will be proportional to the frequency. For example, if one has a >20% delay uncertainty, in a 0.3-second delay, then the phase uncertainty at 10 rad/s, for example, will be >34°. Eq. (G.109) is difficult to use directly in s -plane analysis. Instead, series expansions are employed to represent it. The Taylor-series expansion for an exponential is

$$e^{-sa} = 1 + sa + \frac{(sa)^2}{2} + \frac{(sa)^3}{6} + \frac{(sa)^4}{24} + \frac{(sa)^5}{120} + \dots + \frac{(sa)^n}{n!} \quad (\text{G.111})$$

The transfer function of the exponential should be proper, that is to say the denominator order must be greater than or equal to the order of the numerator. The general approach to expanding Eq. (G.109) is to write it as a fraction

$$e^{-s\Delta} = \frac{e^{-sf\Delta}}{e^{s(1+f)\Delta}} \quad (\text{G.112})$$

where f is less than 1. If f is zero, for example, the first-order expansion is

$$e^{-s\Delta} = \frac{1}{1 + s\Delta} \quad (\text{G.113})$$

This function will represent the exponential at very low frequencies, but does not capture its minimum-phase nature. A widely used representation is with $f = 1/2$. To second order this is

$$e^{-s\Delta} = \frac{1 - \frac{s\Delta}{2} + \frac{(s\Delta)^2}{8}}{1 + \frac{s\Delta}{2} + \frac{(s\Delta)^2}{8}} \quad (\text{G.114})$$

Note that this function has a pair of zeros in the right half-plane and is all-pass. If f and $1-f$ are not equal the expansions in the numerator and denominator should be carried out so that their relative error is equal. For example, if $f = 1/3$ the appropriate function is

$$e^{-s\Delta} = \frac{1 - \frac{s\Delta}{3}}{1 + \frac{2s\Delta}{3} + \frac{(s\Delta)^2}{6}} \quad (\text{G.115})$$

and if $f = 2/5$

$$e^{-s\Delta} = \frac{1 - \frac{2s\Delta}{5} + \frac{(s\Delta)^2}{20}}{1 + \frac{3s\Delta}{5} + \frac{3(s\Delta)^2}{20} + \frac{(s\Delta)^3}{60}} \quad (\text{G.116})$$

where (G.114), (G.115), and Eq. (G.116) are known as Padé approximants.

Example G.8 shows (G.113) (the solid line) and (G.114) (the dot-dashed line) expanded only to first order with $D = 0$ and compared with the exponential (the dashed line). (G.113) rolls off at high frequency and does not match the phase well above 0.2 rad/s. The nonminimum phase-transfer function matches the magnitude exactly and the phase to about 2 rad/s. For frequencies less than $0.1/D$, the delay has minimal effect and can be ignored. Assume that we are designing a rate-control system whose plant is a single integrator with a delay. Represent the delay by (G.114) expanded only to second order. A gain is a perfectly adequate controller for a single integrator. The plant is

$$P(s) = \frac{1 - \frac{s\Delta}{2}}{s(1 + \frac{s\Delta}{2})} \quad (\text{G.117})$$

The controller is K and the closed-loop transfer function is

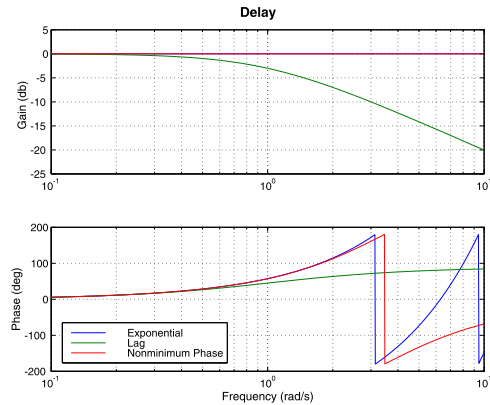
$$\frac{y}{r} = \frac{K(1 - \frac{s\Delta}{2})}{K + \frac{\Delta}{2}s^2 + (1 - K\frac{\Delta}{2})s} \quad (\text{G.118})$$

If $D = 0$ this reduces to the standard equation for rate feedback. The system will become unstable if K exceeds $2/D$. Thus the delay limits the system gain, a natural consequence of a right half-plane zero. Example G.9 shows the Bode magnitude plot with $K = 1$ and $D = 1$. The solid line shows the magnitude response with $D = 0.5$ and the dashed line shows it with $D = 1.0$. The damping decreases as the delay increases. The shape of the magnitude curve is also somewhat different from what you would expect for a

```

1 % Plant
2 clear angle
3 a = [0 1; 0 0]; b = [0; 1];
4 c = [1 0]; d = 0;
5 n = 1000;
6 w = logspace(-1, 1, n);
7 tau = 1;
8 g = exp(1i*w*tau);
9 p = zeros(3, n);
10 m = zeros(3, n);
11 m(1,:) = abs(g);
12 p(1,:) = (180/pi)*angle(g);
13 [num, den] = Pade(0, 1, tau);
14 [m(2,:), p(2,:)] = FResp(num, den, w);
15 [num, den] = Pade(2, 2, tau);
16 [m(3,:), p(3,:)] = FResp(num, den, w);
17 yL = {'Gain_(db)', 'Phase_(deg)'};
18 h = Plot2D(w, [20*log10(m); p], 'Frequency_(rad/s)', ...
19           yL, 'Delay', 'xlog', {'[1:3]', '[4:6]'});
20 legend('Exponential', 'Lag', 'Nonminimum_Phase');
21 set(h, 'color', [1 1 1]);

```

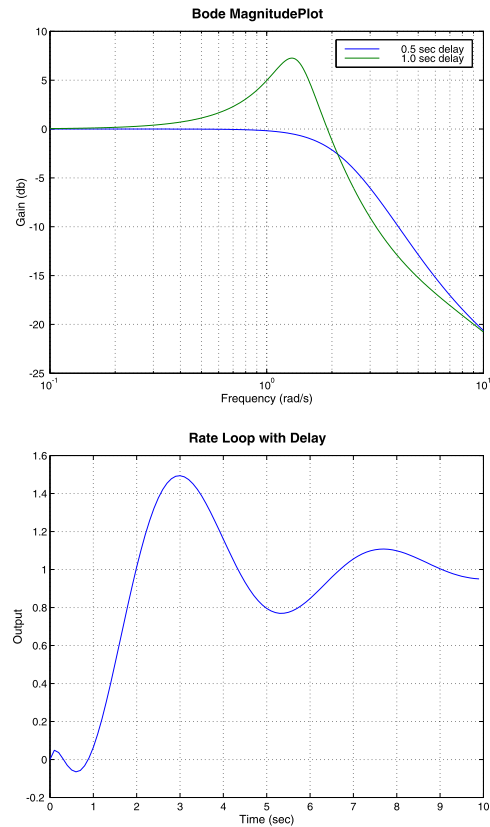


Example G.8: Delay example.

```

1 % Plant
2 aP = 0; bP = 1;
3 cP = 1; dP = 0;
4 w = logspace(-1, 1, 1000);
5 [aD, bD, cD, dD] = Delay(0.5, 2);
6 [a, b, c, d] = Series(aD, bD, cD, dD, aP, bP, cP, dP);
7 [m1, p] = FResp(a-b*c, b, c, d, 1, 1, w);
8 [aD, bD, cD, dD] = Delay(1.0, 2);
9 [a, b, c, d] = Series(aD, bD, cD, dD, aP, bP, cP, dP);
10 [m2, p] = FResp(a-b*c, b, c, d, 1, 1, w);
11 yL = 'Gain_(db)';
12 h = Plot2D(w, 20*log10([m1; m2]), 'Frequency_(rad/s)', ...
13           yL, 'Bode_MagnitudePlot', 'xlog');
14 legend('0.5 sec delay', '1.0 sec delay');
15 g = statespace(a-b*c, b, c, d, 'Rate_Control');
16 [y, x, t] = Step(g, 1, 0.1, 100);
17 Plot2D(t, y, 'Time_(sec)', 'Output', 'Rate_Loop_with_Delay');

```



Example G.9: Bode magnitude plot of the rate-control loop with delays of 0.5 and 1 second.

second-order system with the same damping ratio. At high frequencies, it appears to be concave instead of convex.

The zero has an interesting effect on the transient response. At low frequencies, the transfer function is 1. At high frequencies, the transfer function is

$$\frac{y}{r} = -\frac{K}{s} \quad (\text{G.119})$$

The sign is reversed! Therefore any input (such as a step) with sufficiently high-frequency content will cause the output to go in the opposite direction of the input at first. This is illustrated in the bottom plot of Example G.9. This effect can be reduced by shortening the delay or low-pass filtering the command input.

G.9.2.4 The zero-order hold

The zero-order hold is the standard method of driving analog actuators with signals from computers. The zero-order hold is given by

$$u(t) = u_{-1}(t) - u_{-1}(t - T) \quad (\text{G.120})$$

This is just the difference of a step and a delayed step. The Laplace transform of Eq. (G.113) is

$$u(s) = \frac{1 - e^{-sT}}{s} \quad (\text{G.121})$$

When multiplied by the gain of the sampler, $1/T$, it becomes

$$u(s) = \frac{1 - e^{-sT}}{sT} \quad (\text{G.122})$$

Eq. (G.122), when included in the continuous-time system, represents the effects of sampling on input and the zero-order hold on output. This allows us to include the effects of the digital system while designing in the continuous-time domain. Eq. (G.122) is most easily handled using its series expansion. This is done by substituting Eq. (G.115) or Eq. (G.116) for the delay in the numerator. The result is

$$u(s) = \frac{1 - \frac{sT}{6}}{1 + \frac{sT}{3}} \quad (\text{G.123})$$

for Eq. (G.115) and

$$u(s) = \frac{1 - \frac{sT}{10} + \frac{(sT)^2}{60}}{1 + \frac{2sT}{5} + \frac{(sT)^2}{20}} \quad (\text{G.124})$$

for Eq. (G.116). Substituting Eq. (G.114), would not have provided the right half-plane zeros that give the zero-order hold its nonminimum phase character.

G.9.2.5 Pulswidth modulation

Pulswidth modulation involves varying the pulswidth from 0 to T . While this cannot be directly represented, as can the zero-order hold, the effect can be analyzed by assuming that the pulswidth is fixed at t , a fraction of T . The pulswidth modulator is

$$u(t) = u_{-1}(t - T + \tau) - u_{-1}(t - T) \quad (\text{G.125})$$

The Laplace transform (including the sampler gain) is

$$u(s) = \frac{e^{-s(T-\tau)} - e^{-sT}}{sT} \quad (\text{G.126})$$

or, after some manipulation

$$u(s) = \left(\frac{e^{s\tau} - 1}{e^{sT} - 1} \right) \left(\frac{1 - e^{sT}}{sT} \right) \quad (\text{G.127})$$

The first term in parentheses is an additional nonminimum-phase function. Using

$$e^{sT} - 1 = \frac{1 + \frac{sT}{10} + \frac{(sT)^2}{60}}{1 - \frac{2sT}{5} + \frac{(sT)^2}{20}} \frac{1}{s} \quad (\text{G.128})$$

and substituting gives

$$\frac{e^{s\tau} - 1}{e^{sT} - 1} = \frac{\tau}{T} \left(\frac{1 + \frac{s\tau}{10} + \frac{(s\tau)^2}{60}}{1 - \frac{2s\tau}{5} + \frac{(s\tau)^2}{20}} \right) \left(\frac{1 - \frac{2sT}{5} + \frac{(sT)^2}{20}}{1 + \frac{sT}{5} + \frac{(sT)^2}{60}} \right) \quad (\text{G.129})$$

The finite pulswidth introduces a very high-frequency zero pair and an unstable pole pair. For frequencies below the sampling frequency, determined by T , this transfer function does not appreciably affect the dynamics. Therefore the transfer function for the pulswidth modulator is

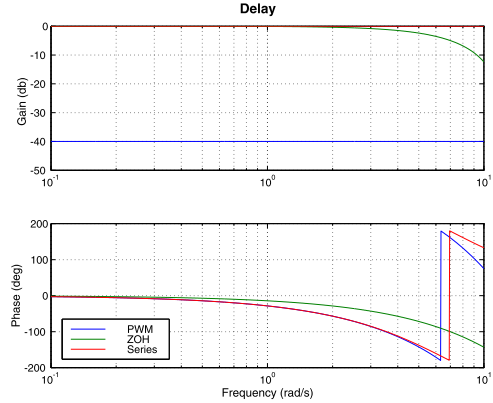
$$u(s) = \frac{\tau}{T} \left(\frac{11 \frac{2sT}{5} + \frac{(sT)^2}{20}}{1 + \frac{sT}{5} + \frac{(sT)^2}{60}} \right) \left(\frac{1 - \frac{sT}{10} + \frac{(sT)^2}{60}}{1 + \frac{2sT}{5} + \frac{(sT)^2}{20}} \right) \quad (\text{G.130})$$

Example G.10 shows the Bode plot for the exact and series expansion, as well as the zero-order hold, for $T = 0.5$ and $t = 0.005$. The solid line is the series expansion for the pulswidth modulator, the dashed line is the exact representation, and the dot-dashed line is the zero-order hold. The gain on the zero-order hold was made identical to that of the pulswidth modulator for comparison purposes only. The pulswidth modulator has a greater phase shift than the zero-order hold and no high-frequency roll-off. It will tend to interact more with high-frequency dynamics.

```

1 w = logspace(-1,1,1000);
2 tau = 0.005;
3 T = 0.5;
4 s = j*w;
5 % PWM
6 g = (exp(-s*(T-tau)) - exp(-s*T)) ./ (s*T);
7 m(1,:) = abs(g);
8 p(1,:) = (180/pi)*angle(g);
9 % ZOH
10 g = (1-exp(-s*T)) ./ (s*T);
11 m(2,:) = abs(g);
12 p(2,:) = (180/pi)*angle(g);
13 [num,den] = PWM(tau, T, 2);
14 [m(3,:),p(3,:)] = FResp(num,den,w);
15 yL = {'Gain_(db)', 'Phase_(deg)'};
16 h = Plot2D(w, [20*log10(m);p], ...
17             'Frequency_(rad/s)', ...
18             yL, 'Delay', 'xlog', ...
19             {'[1:3]', '[4:6]'});
20 legend('PWM', 'ZOH', 'Series');
21 set(h, 'color',[1 1 1]);

```



Example G.10: Comparison of pulsewidth modulator and zero-order hold.

G.10. Continuous-to-discrete transformations

G.10.1 The difference equation

Once we have a continuous design, it is necessary to convert it into a set of difference equations for implementation in the computer. The controller will be of the form

$$\zeta_i = \sum_{j=i-1}^{i-n} a_j x_j + \sum_{j=i}^{i-m} b_j u_j \quad (\text{G.131})$$

where u_j is the j th input and x_j is the j th output. The time between inputs and outputs is the sample period T . If we define z as a one sample-period delay, we can transform the difference equation into

$$x(z) = \sum_{j=1}^n a_j z^j x(z) + \sum_{j=0}^m b_j z^j u(z) \quad (\text{G.132})$$

For example, the difference equation for a trapezoidal rule integrator is

$$x_i = x_{i-1} + \frac{T}{2}(u_i + u_{i-1}) \quad (\text{G.133})$$

When it has $n = 1$ and $m = 1$ and is transformed into the z domain it is

$$\frac{x(z)}{u(z)} = \frac{T}{2} \frac{z+1}{z-1} \quad (\text{G.134})$$

G.10.2 Transforming from the s plane to the z plane

The delay z in the s plane is

$$z = e^{sT} \quad (\text{G.135})$$

and

$$s = \frac{\log z}{T} \quad (\text{G.136})$$

Eq. (G.135) and Eq. (G.136) allow us to move between the s and z planes. Given a continuous time-transfer function, the equivalent s domain function can be found by substituting Eq. (G.136) for s . Since the natural logarithm does not easily convert to difference equations, we expand it in a series.

$$s = \frac{2}{T} \left(\left(\frac{z-1}{z+1} \right) + \frac{1}{3} \left(\frac{z-1}{z+1} \right)^3 + \dots + \frac{1}{n} \left(\frac{z-1}{z+1} \right)^n \right) \quad (\text{G.137})$$

where n is odd. The first-order expansion is known as the Tustin transformation and is

$$s = \frac{2}{T} \left(\frac{z-1}{z+1} \right) \quad (\text{G.138})$$

The second-order expansion is

$$s = \frac{8}{3T} \left(\frac{z^3-1}{(z+1)^3} \right) \quad (\text{G.139})$$

Since s is the derivative operator, one could also replace it with a first difference

$$s = \frac{1}{T} (z - 1) \quad (\text{G.140})$$

or a second difference

$$s = \frac{1}{T} (3z^2 - 4z + 1) \quad (\text{G.141})$$

G.10.3 Transformation of a differentiator

A differentiator with roll-off at high frequencies is

$$\frac{s}{\tau s + 1} \quad (\text{G.142})$$

The response to a ramp at $t = \infty$ can be found by the final-value theorem

$$\lim_{t \rightarrow \infty} = \lim_{s \rightarrow 0} s \left(\frac{s}{\tau s + 1} \right) \frac{1}{s^2} = 1 \quad (\text{G.143})$$

This can be implemented with a first difference. The result is

$$\frac{1}{\tau} \left(\frac{z-1}{z - (1 - \frac{T}{\tau})} \right) \quad (\text{G.144})$$

where $\tau > T$. The final-value theorem can be applied to this function

$$\lim_{t \rightarrow \infty} = \lim_{s \rightarrow 0} s \left(\frac{e^{sT} - 1}{e^{sT} - (1 - \frac{T}{\tau})} \right) \frac{1}{s^2} = \frac{1}{T} \frac{e^{sT} - 1}{s} = \frac{sT}{sT} = 1 \quad (\text{G.145})$$

The matched pole-zero method is also easy to apply. Transforming the zero and pole gives

$$K \frac{z-1}{z - e^{T/\tau}} \quad (\text{G.146})$$

The gain is found using the final-value theorem

$$K \lim_{s \rightarrow 0} \left(\frac{1}{e^{sT} - e^{T/\tau}} \right) \frac{e^{sT} - 1}{s} = 1 \quad (\text{G.147})$$

or

$$K = \frac{1 - e^{T/\tau}}{T} \quad (\text{G.148})$$

These results, along with the Tustin and the 2nd difference are summarized in Table G.6.

All four rate-estimator errors are shown below.

G.10.4 State estimator

An alternative is to use a state estimator. The discrete-time state equations for a double-integrator plant are

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}_{k+1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_k + \begin{bmatrix} \frac{T^2}{2} \\ T \end{bmatrix} u + \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \left(\gamma - \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_k \right) \quad (\text{G.149})$$

The first state is the estimated angle and the second is the estimated rate. γ is the measured angle. Expanding the state equations as difference equations gives

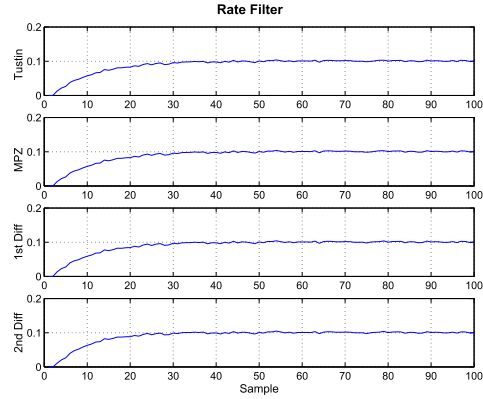
$$zx_1 = x_1 + Tx_2 + k_1 - k_1x_1 \quad (\text{G.150})$$

$$zx_2 = x_2 + k_2\gamma - k_2x_1 \quad (\text{G.151})$$

```

1 T      = 0.1;
2 tau    = 1;
3 % Tustin
4 [nT, dT] = S2Z( [1 0], [tau 1], T, ...
5               'bilinear' );
6 % MPZ
7 k      = (1 - exp(-T/tau))/T;
8 nM     = k*[1 -1];
9 dM     = [1 -exp(-T/tau)];
10 % First difference
11 nF     = (1/tau)*[1 -1];
12 dF     = [1 -(1-T/tau)];
13 % 2nd difference
14 nS     = [3 -4 1]/tau/3;
15 dS     = [3 -4 1 + 2*T/tau]/3;
16 % Sim
17 nSim   = 100;
18 y      = zeros(4, nSim);
19 u      = linspace(0,1,nSim) + 0.002*randn(1,
20               nSim);
21 for k = 3:nSim
22     y(1,k) = -dT(2)*y(1,k-1) + nT(1)*u(k) ...
23             + nT(2)*u(k-1);
24     y(2,k) = -dM(2)*y(2,k-1) + nM(1)*u(k) ...
25             + nM(2)*u(k-1);
26     y(3,k) = -dF(2)*y(3,k-1) + nF(1)*u(k) ...
27             + nF(2)*u(k-1);
28     y(4,k) = -dS(2)*y(4,k-1) ...
29             - dS(3)*y(4,k-2) ...
30             + nS(1)*u(k) ...
31             + nS(2)*u(k-1) ...
32             + nS(3)*u(k-2);
33 end
34 yL = ['Tustin' 'MPZ' '1stDiff' '2ndDiff'];
35 h = Plot2D( 1:nSim, y, 'Sample', yL, 'RateFilter' );
36 set(h, 'color', [1 1 1]);
37 PrintFig(0,1,1, 'RateEst')
38 err = mean(y(:,(nSim/2):end) - 0.1,2)
39 %err =
40 % 0.00096175387563
41 % 0.00096088706325
42 % 0.00100900220493
43 % 0.00110102280894

```



Example G.11: Comparison of the four rate estimators.

Solving the first equations for x_1 gives

$$x_1 = \frac{Tx_2 + k_1 y}{z - 1 + k_1} \quad (\text{G.152})$$

and substituting into the second equation for x_1 gives the transfer function

$$\frac{x_2(z)}{y(z)} = k_2 \frac{z - 1}{z^2 + (k_1 - 2)z + Tk_2 - k_1 + 1} \quad (\text{G.153})$$

which has a second-order denominator. The gains can be chosen in many ways. One is just to map the desired s -plane poles into the z -plane using the relationship (see Table G.6)

$$z = e^{sT} \quad (\text{G.154})$$

Table G.6 Z-plane representation of a filtered differentiator.

Form	Transfer Function
First Difference	$\frac{1}{\tau} \left(\frac{z-1}{z - \left(1 - \frac{T}{\tau}\right)} \right)$
Second Difference	$\frac{1}{\tau} \left(\frac{3z^2 - 4z + 1}{3z^2 - 4z + 1 + \frac{2T}{\tau}} \right)$
Tustin	$\frac{2/\tau}{1 + \frac{T}{2\tau}} \frac{z-1}{z - \left(\frac{1 - \frac{T}{2\tau}}{1 + \frac{T}{2\tau}}\right)}$
Matched Pole Zero	$\frac{1 - e^{-T/\tau}}{T} \left(\frac{z-1}{z - e^{-T/\tau}} \right)$
State Estimator	$\frac{x_2(z)}{y(z)} = k_2 \frac{z-1}{z^2 + (k_1 - 2)z + Tk_2 - k_1 + 1}$

where

$$s = -\zeta\omega \pm j\omega\sqrt{1 - \zeta^2} \quad (\text{G.155})$$

so

$$z = e^{-\zeta\omega T} e^{\pm j\theta} \quad (\text{G.156})$$

Note that

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (\text{G.157})$$

so the roots are

$$z = e^{-\zeta\omega T} \begin{bmatrix} \cos \theta + j \sin \theta & \cos \theta - j \sin \theta \end{bmatrix} \quad (\text{G.158})$$

This gives

$$\theta = \omega T \sqrt{1 - \zeta^2} \quad (\text{G.159})$$

$$k_1 = 2(e^{-\zeta\omega T} \cos \theta + 1) \quad (\text{G.160})$$

$$k_2 = \frac{e^{-\zeta\omega T} (1 + 2 \cos \theta) + 1}{T} \quad (\text{G.161})$$

where

$$\theta = \omega \sqrt{1 - \zeta^2} \quad (\text{G.162})$$

The roots of the transfer function in z are

$$z = -\frac{k_1 - 2}{2} \pm \sqrt{\frac{(k_1 - 2)^2}{4} - Tk_2 + k_1 - 1} \quad (\text{G.163})$$

Solving for k_1 and k_2

$$k_1 = 2(1 - e^{-\zeta\omega T} \cos \theta) \quad (\text{G.164})$$

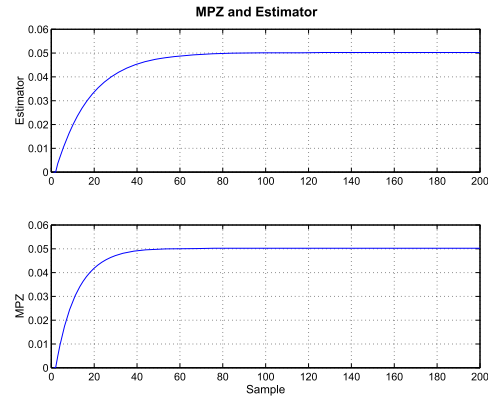
$$k_2 = \frac{1}{T} \left(\frac{(k_1 - 2)^2}{4} + k_1 - 1 - \sin^2 \theta e^{-2\zeta\omega T} \right) \quad (\text{G.165})$$

The s -plane parameters for the estimator are $z = 0.7072$ and $\omega = 10.0$.

```

1 T = 0.1;
2 tau = 1;
3 zeta = 0.7071;
4 omega = 10.0;
5 % MPZ
6 k = (1 - exp(-T/tau))/T;
7 nM = k*[1 -1];
8 dM = [1 -exp(-T/tau)];
9 % Estimator
10 a = [1 T; 0 1];
11 theta = omega*T*sqrt(1-zeta^2);
12 c = cos(theta);
13 e = exp(-zeta*omega*T);
14 k1 = 2*(1 - e*c);
15 s = sin(theta);
16 k2 = ((k1-2)^2/4 + k1 - 1 - e*s)/T;
17 K = [k1; k2];
18 c1 = (K(1)-2);
19 c0 = T*K(2)-(K(1)+1);
20 % Sim
21 nSim = 200;
22 y = zeros(2, nSim);
23 u = linspace(0, 1, nSim) + 0.00*randn(1, nSim);
24 x = zeros(2, 1);
25 for k = 3:nSim
26     y(1, k) = K(2)*(u(k) - u(k-1)) ...
27             - c1*y(1, k-1) - c0*y(1, k-2);
28     %x = a*x + K*(u(k)-x(1));
29     y(2, k) = -dM(2)*y(2, k-1) + nM(1)*u(k) ...
30             + nM(2)*u(k-1);
31 end
32 yL = {'Estimator' 'MPZ'};
33 h = Plot2D(1:nSim, y, 'Sample', yL, ...
34           'MPZ_and_Estimator');
35 set(h, 'color', [1 1 1]);
36 PrintFig(0, 1, 1, 'MPZAndEstimator')

```



Example G.12: Comparison of the matched pole zero with the estimator.

The Tustin and first difference are just low-order expansions of the matched pole zero. The second difference has zeros at 1 and $1/3$ and poles at

$$\frac{4 \pm \sqrt{4 - \frac{24T}{\tau}}}{6} \quad (\text{G.166})$$

G.11. Flexible-structure control

G.11.1 Introduction

Most vehicles cannot be considered rigid bodies when their control systems are being designed. Generally, a flexible body is one in which the structure deforms and those deformations are much smaller than the rigid-body motion. However, many other types of dynamics can be included in this category.

G.11.2 Two coupled inertias

The simplest possible flexible-body problem is that of two coupled inertias, illustrated in Fig. G.9. The inertias are connected by a torsional spring and damper. A control torque can be applied to body 1. Disturbances may be applied to either body. The linear differential equations for the coupled set are

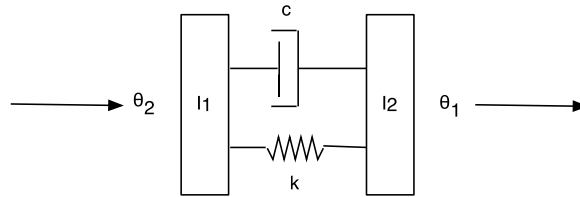


Figure G.9 Two inertias coupled by a spring and a damper.

$$I_1 \ddot{\theta}_1 + c(\dot{\theta}_1 - \dot{\theta}_2) + k(\theta_1 - \theta_2) = u + d_1 \quad (\text{G.167})$$

$$I_2 \ddot{\theta}_2 + c(\dot{\theta}_2 - \dot{\theta}_1) + k(\theta_2 - \theta_1) = d_2 \quad (\text{G.168})$$

Taking the Laplace transform of the equations gives

$$\begin{bmatrix} I_1 s^2 + cs + k & -cs - k \\ -cs - k & I_2 s^2 + cs + k \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} u + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \quad (\text{G.169})$$

and solving for θ_1 and θ_2 gives

$$\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \frac{\begin{bmatrix} I_1 s^2 + cs + k & -cs - k \\ -cs - k & I_2 s^2 + cs + k \end{bmatrix}}{I_1 I_2 s^2 (s^2 + c \frac{\gamma}{I_2} s + k \frac{\gamma}{I_2})} \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \quad (\text{G.170})$$

where γ is the modal coupling factor and is

$$\gamma = \frac{I_1 + I_2}{I_1} \quad (\text{G.171})$$

and γ can range from 1 to ∞ .

G.11.3 Double integrator

The collocated plant reduces to a double integrator if $\gamma \rightarrow 1$. The double integrator can be controlled by a simple lead network.

$$C(s) = K \frac{\tau_1 s + 1}{\tau_2 s + 1} \quad (\text{G.172})$$

where $\tau_1 > \tau_2$. That is, the zero is at a lower frequency than the pole. This can also be thought of as a proportional-derivative controller cascaded with a low-pass filter. The advantage of using the pole/zero combination is that the phase-lag effects of the pole are automatically accounted for in the design.

First, choose the crossover frequency and create a lead network that provides a + phase shift at that frequency and then adjust the gain so the crossover (where the gain is 1) is at that frequency. This is shown in the following example. The crossover is set to 0.05 rad/s. 60° of lead is selected. In the first example the damping ratios for the complex modes are low. When the gain is lowered by 63 dB (so now the crossover is centered in the lead peak) the damping ratios are all 1.

G.11.4 Control algorithms

Two cases are of interest. The first is the collocated sensor and actuator case. The transfer function from u to θ_1 is

$$\frac{\theta_1}{u} = \frac{s^2 + \frac{c}{I_2}s + \frac{k}{I_2}}{I_1 s^2 (s^2 + c \frac{\gamma}{I_2} s + k \frac{\gamma}{I_2})} \quad (\text{G.173})$$

When $I_1 \gg I_2$ the modal coupling factor is 1 and the quadratic factor in the denominator equals the quadratic factor in the numerator and they drop out. The resulting transfer function is for a rigid body with inertia I_1 .

At frequencies much lower than the roots of the denominator quadratic factor, the first two terms in the quadratic factor are negligible and the transfer function reduces to a rigid body with inertia $I_1 + I_2$. At frequencies much higher than the complex poles, the transfer function reduces to a rigid body with inertia I_1 .

Since γ is always greater than 1, the pole pair determined by the quadratic term in the denominator will always be at a higher frequency than the zeros. Therefore the phase that starts at -180 deg due to the double pole at the origin, will flip to 0° and then return to -180°. No matter how many flexible-body modes there are, if the sensor and actuator are collocated, the phase is never less than -180°.

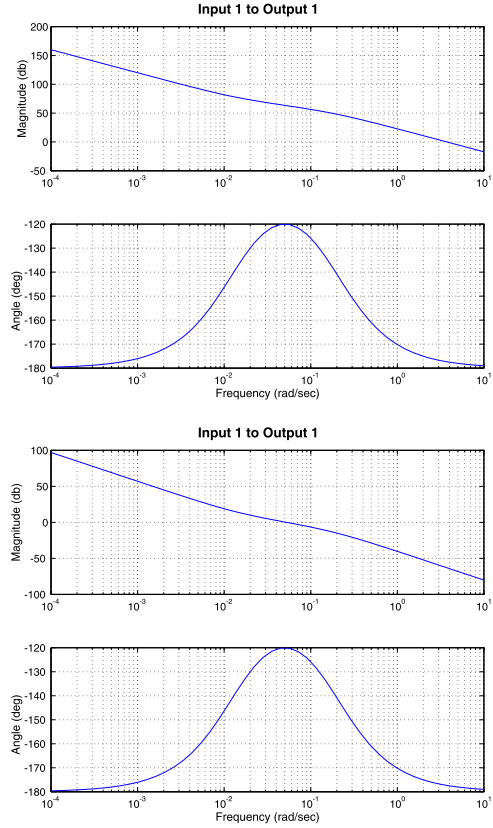
The damping ratio for the fixed-base mode is

$$\zeta = \frac{c}{s\sqrt{kI_2}} \quad (\text{G.174})$$

```

1 % Double integrator
2 [aP, bP, cP, dP] = ND2SS( 1, [1 0 0] );
3 % Lead
4 [aL, bL, cL, dL] = LeadLag( 0.05, 60, 0 );
5 [a, b, c, d] = Series(aL, bL, cL, dL, aP, bP, cP, dP);
6 FResp(a, b, c, d);
7 s = eig(a-b*c)
8 zeta = S2Damp(s)
9 % s =
10 %
11 % -0.0134
12 % -0.0866 + 3.7307i
13 % -0.0866 - 3.7307i
14 %
15 %
16 % zeta =
17 %
18 % 1.0000
19 % 0.0232
20 % 0.0232
21 % With gain adjustment
22 [aL, bL, cL, dL] = LeadLag( 0.05, 60, -63 );
23 [a, b, c, d] = Series(aL, bL, cL, dL, aP, bP, cP, dP);
24 FResp(a, b, c, d);
25 s = eig(a-b*c)
26 zeta = S2Damp(s)

```



Example G.13: Control of a double integrator with a lead network.

and the damping ratio for the moving-base case is

$$\zeta = \frac{c}{s\sqrt{\gamma k I_2}} \quad (\text{G.175})$$

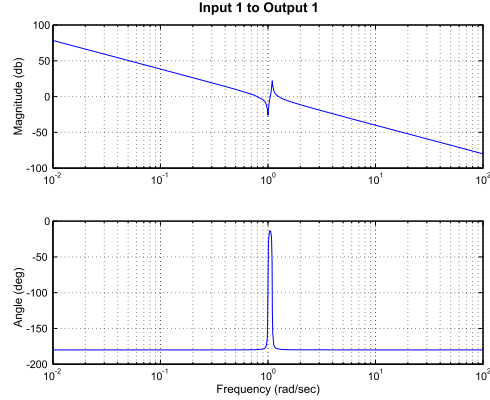
Since γ is always greater than one, the damping ratio will always be less than that measured on the ground. The poles and zeros of the quadratic terms are always complex since flexible modes (if they are of concern at all) are lightly damped. The numerator zeros are the fixed-base frequencies of the flexible part of the system (represented by I_2 and its spring and damper). They are the frequencies that would be measured if the flexible part were cantilevered to a fixed base. It is easy to go from the fixed-base vibrational frequencies to the frequencies of the vehicle if one knows the inertias.

The Bode plot for this transfer function for $I_1 = 1$, $I_2 = 0.1$, fixed base $\zeta = 0.005$ and fixed-base frequency equal to 1 rad/s is illustrated in Example G.14.

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [1 c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 FResp(num,den)
11 PrintFig(1,1,1,'Collocated')

```



Example G.14: Bode plot for the collocated sensor and actuator transfer function.

Note that this system has multiple crossovers. This is a characteristic of flexible-body systems that is not seen in many other control problems.

Suppose that instead of measuring θ_1 , we measure θ_2 . θ_2 is separated from the actuator by the flexible coupling. The transfer function is

$$\frac{\theta_2}{u} = \frac{\frac{c}{I_2}s + \frac{k}{I_2}}{I_1 s^2 (s^2 + c \frac{\gamma}{I_2} s + k \frac{\gamma}{I_2})} \quad (\text{G.176})$$

The numerator is a single real zero at frequency k/c . In terms of damping ratio this is

$$s = \frac{\omega}{2\zeta} \quad (\text{G.177})$$

If ζ is small, this is much higher than the denominator frequency. Therefore in the noncollocated system, the phase will decrease to -360° after the pole pair and will not return to -180° until a much higher frequency.

Note that the system behavior at very high and very low frequencies (relative to the complex) pole is not changed. It is only the behavior at intermediate frequencies that is different.

The Bode plot for the noncollocated transfer function is illustrated in Example G.15. The transfer functions for the different frequency domains for both collocated and noncollocated transfer functions are summarized in Table G.7. At intermediate frequencies, the motion is governed by the flexible-body modes. At very high frequencies only the core body moves.

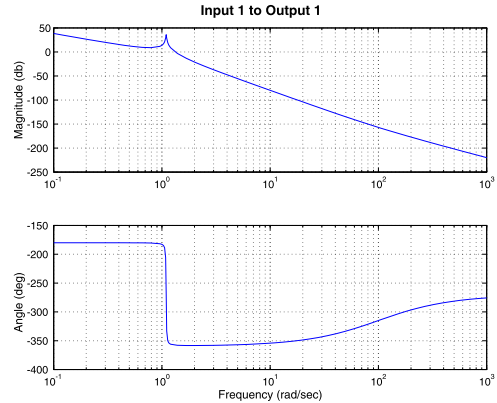
G.11.5 Lead compensation of the minimum-phase system

Our goal is to provide a well-damped response to an input command. Therefore it is required to add damping to the double pole at the origin and to add some damping to

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 FResp(num,den)

```



Example G.15: Bode plot for the noncollocated sensor and actuator transfer function.

Table G.7 Transfer-function summary.

Frequency	Collocated	Noncollocated
Low	$\frac{1}{(I_1 + I_2)s^2}$	$\frac{1}{(I_1 + I_2)s^2}$
Intermediate	$\frac{s^2 + \frac{c}{I_2}s + \frac{k}{I_2}}{I_1 s^2 (s^2 + c\frac{\gamma}{I_2}s + k\frac{\gamma}{I_2})}$	$\frac{1}{I_1 s^2 (s^2 + c\frac{\gamma}{I_2}s + k\frac{\gamma}{I_2})}$
High	$\frac{1}{I_1 s^2}$	$\frac{1}{I_1 s^2}$

the flexible mode. We have three degrees-of-freedom in the lead compensator. We can adjust the gain, we can choose the maximum amount of phase lead it can add, and we can select the frequency where the phase lead is at its maximum.

If we were not concerned with damping the flex mode, we would put our maximum phase shift at the gain crossover. However, if we want to dampen the flex modes, we want to add some phase lead where they have their gain crossovers. There are three gain crossovers, one at a frequency lower than the resonance and two associated with the resonance pole.

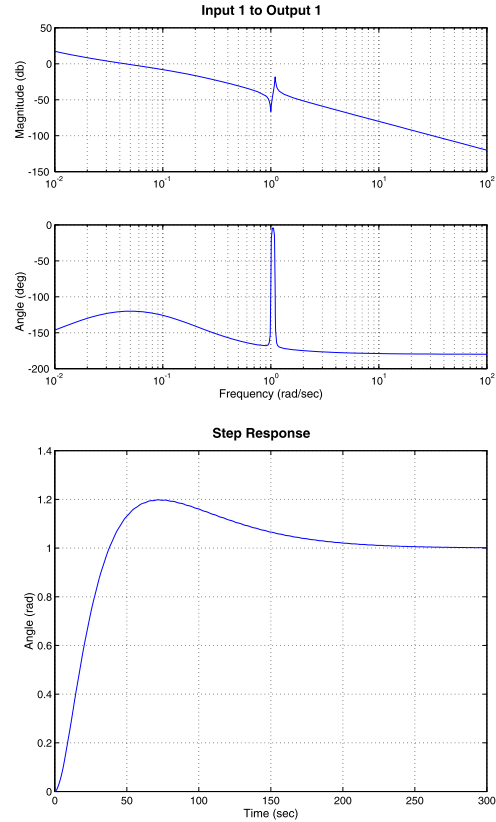
The first step is just to introduce our flex mode into the system for which we have designed the controller above. The result is that the damping at the crossover changes little but the flex-mode damping is still very light. We have not done anything to suppress the flex mode. If we push the lead up to 0.5 rad per second we can add phase lead to the flex mode.

This simple compensator is not the best compensator that could be designed for this system. However, it serves to demonstrate what can be achieved with a very simple design approach. Even with this compensator, the four transient responses do not

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [1 c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 [aP, bP, cP, dP] = ND2SS( num, den );
11 % Lead
12 [aL, bL, cL, dL] = LeadLag( 0.05, 60, -63 );
13 [a,b,c,d] = Series(aL,bL,cL,dL,aP,bP,cP,dP);
14 FResp(a,b,c,d);
15
16 s = eig(a-b*c)
17 zeta = S2Damp(s)
18 g = statespace(a-b*c,b,c,d);
19 dT = 0.1;
20 nSim = 3000;
21 y = Step( g, 1, dT, nSim );
22 t = (0:(nSim-1))*dT;
23 Plot2D( t, y, 'Time(sec)', 'Angle(rad)', 'Step
    Response' );

```



Example G.16: Lead compensator with flex node added.

necessarily show the optimal response. One would not leave the maximum phase lead frequency fixed as the gain and crossover were increased. Rather, one would push the lead frequency up. This would eliminate the oscillatory transient response, but would not increase the damping on the flex mode. To push the bandwidth up and increase the damping on the flex mode, a different compensation approach would be required.

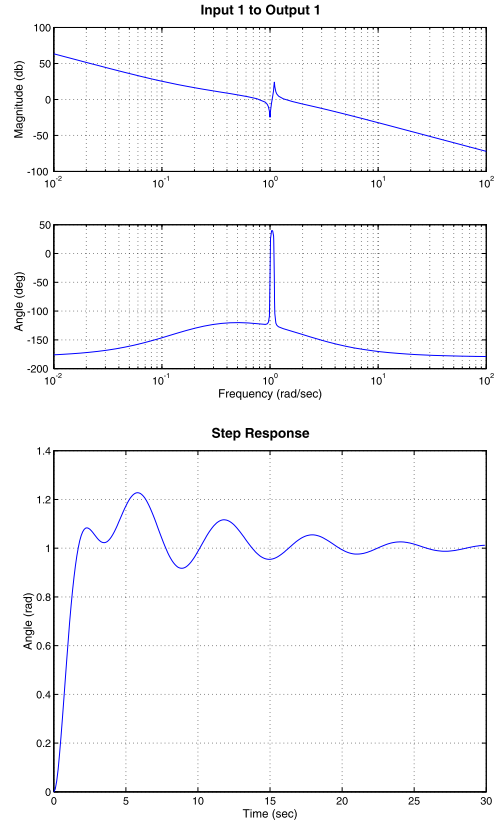
G.11.6 Noncollocated sensor and actuator

Example G.18 shows the Bode plot for a noncollocated sensor and actuator. Example G.18 shows the frequency response over a wider range to show the effect of the high-frequency zero. As expected, it reduces the phase lag to -270° at high frequencies. Clearly, at frequencies below the resonance, the control problem is the same as for the collocated sensor and actuator. To show this, use a phase-lead controller with a crossover at 0.05 rad/s. The Bode plot is shown in Example G.19. -62 dB was chosen for the forward gain. The lead introduced by the controller stabilizes both the control crossover at

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [1 c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 [aP, bP, cP, dP] = ND2SS( num, den );
11 % Lead
12 [aL, bL, cL, dL] = LeadLag( 0.5, 60, -15 );
13 [a,b,c,d] = Series(aL,bL,cL,dL,aP,bP,cP,dP
);
14 FResp(a,b,c,d);
15
16 s = eig(a-b*c)
17 zeta = S2Damp(s)
18 g = statespace(a-b*c,b,c,d);
19 dT = 0.1;
20 nSim = 300;
21 y = Step( g, 1, dT, nSim );
22 t = (0:(nSim-1))*dT;
23 Plot2D( t, y, 'Time(sec)', 'Angle(rad)', 'Step
Response' );

```



Example G.17: Lead compensator providing flex damping.

0.05 rad/s and the first resonance crossover. The second crossover is stable because the gain is less than 0 dB. This controller has about 16 dB of margin. Example G.20 shows the root-locus plot. The system has two zeros at infinity, a high-frequency plant zero and the controller zero. The two rigid-body poles at the origin go around the controller zero, coalesce on the real axis then one goes to the plant zero and the other goes to the controller zero. The two flex poles go to the zeros at infinity using the right half-plane. They cross into the right half-plane when the gain is about -46 dB.

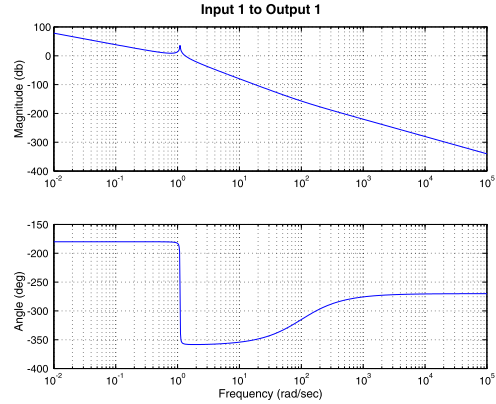
The controller, even at low gains, does not do anything to improve the damping on the flex mode. At best, it ignores it. Unlike the collocated case, the flex poles do not conveniently approach the open-loop zeros. Instead, they go unstable.

Ignoring the flex mode may be all that is needed to meet specifications. If that is the case the simple controller will do. Suppose, however, that you want your crossover to be above 1 rad/s? The simple lead compensator will not work since it is only supplying a maximum of 60° of phase lead. To compensate this plant, we need a pair of zeros

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 w = logspace(-2,5,1000);
11 FResp(num,den,w)

```

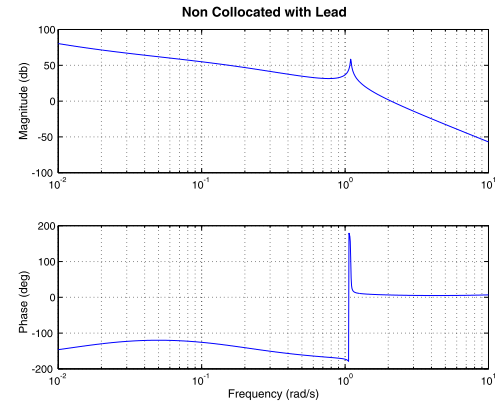


Example G.18: Bode plot for the noncollocated sensor and actuator transfer function.

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 [aP, bP, cP, dP] = ND2SS( num, den );
11 wLead = 0.05;
12 [aL, bL, cL, dL] = LeadLag( wLead, 60, 0 );
13 [a, b, c, d] = Series( aL, bL, cL, dL, aP, bP, cP, dP );
14 w = logspace(-2,1,1000);
15 [m,p] = FResp(a,b,c,d,1,w);
16 yL = { 'Magnitude(db)' 'Phase(deg)' };
17 Plot2D(w,[20*log10(m);p],'Frequency(rad/s)',yL,'
    NonCollocated with Lead');

```



Example G.19: Bode plot for the noncollocated sensor and actuator transfer function with a phase-lead controller and a crossover at 0.05 rad/s.

to attract the flex poles. However, every pair of zeros requires another pair of poles, otherwise, the controller would not be proper. In this plant just adding a pair of zeros (in addition to the lead compensator) would mean that at high frequencies the plant rolls off like $1/s$, which is inadequate when unmodeled dynamics are considered. The flex compensator will have the form

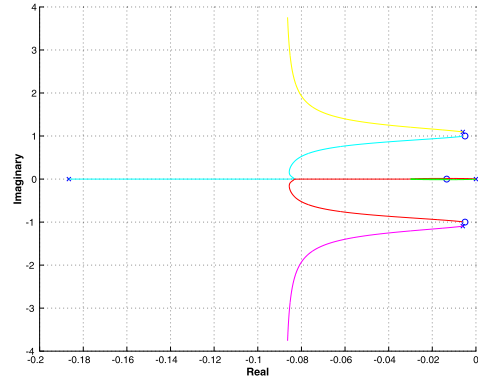
$$\frac{s^2 + 2\xi_1\omega_1s + \omega_1^2}{s^2 + 2\xi_2\omega_2s + \omega_2^2} \quad (\text{G.178})$$

Start by putting a lead compensator with a crossover at 2 rad/s. As expected, the resonance is now unstable since at the phase crossover the gain is more than 1. We need

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [1 c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 [aP, bP, cP, dP] = ND2SS( num, den );
11
12 [aL, bL, cL, dL] = LeadLag( 0.05, 60, 0 );
13 [a,b,c,d] = Series(aL,bL,cL,dL,aP,bP,cP,dP);
14 g = statespace(a,b,c,d);
15 RootLocus( g, logspace(-8,0,10000) );

```



Example G.20: Root-locus plot for the noncollocated sensor and actuator transfer function with a phase-lead controller.

to advance the phase by nearly 180° in the vicinity of the resonance, and the controller crossover.

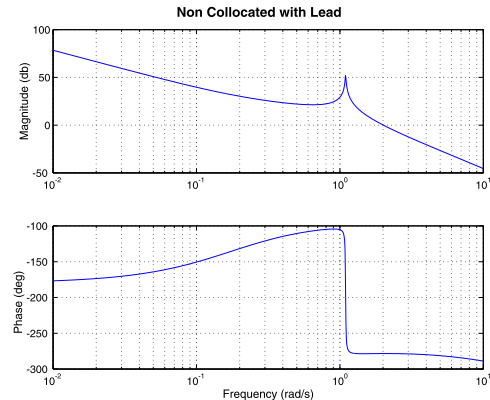
To get the proper phase advance requires that the numerator of Eq. (G.178) be at a lower frequency than the denominator. The zeros should also be at a lower frequency than the resonance to get the full effect of the zeros before hitting the resonant poles. The appropriate pole and zero frequencies can be achieved with poles or zeros in either the right half-plane or the left half-plane.

Example G.21 shows the crossover at 2 rad/s. The system is unstable due to the phase advance of the zero. Physically this means that we are trying to move the system faster than we can read the response across the flexible connection.

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [1 c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 [aP, bP, cP, dP] = ND2SS( num, den );
11 wLead = 2;
12 [aL, bL, cL, dL] = LeadLag( wLead, 80, 0 );
13 [a,b,c,d] = Series(aL,bL,cL,dL,aP,bP,cP,dP);
14 w = logspace(-2,1,1000);
15 [m,p] = FResp(a,b,c,d,1,w);
16 yL = {'Magnitude_(db)' 'Phase_(deg)'};
17 j = find(p >= 0);
18 p(j) = p(j) - 360;
19 Plot2D(w,[20*log10(m);p], 'Frequency_(rad/s)', yL, 'Non Collocated with Lead');

```



Example G.21: Bode plot with a high-frequency phase-lead controller.

The question of where to put the zeros has to do with our knowledge of the system. Try two different flex compensators of the form

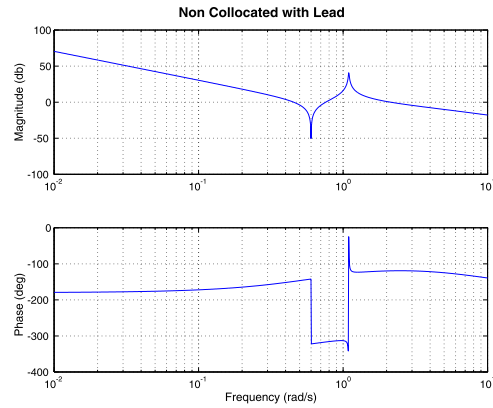
$$\frac{s^2 + (1 \pm 0.4)^2}{s^2 + 0.2 \times 30 + 30^2} \quad (\text{G.179})$$

The damping in the denominator is necessary to stabilize the high-frequency pole introduced by the compensator. This pole is necessary for the compensator to be implementable, you cannot just add the zeros by themselves. The controller pole is high frequency and will damp quickly even with a modest damping ratio. Example G.22 shows that the 2nd-order transfer function provides enough phase shift to result in a stable compensator.

```

1 I1 = 1;
2 I2 = 0.2;
3 zeta = 0.005;
4 omega = 1;
5 k = I2*omega^2;
6 gamma = (I1 + I2)/I1;
7 c = 2*zeta*omega*I2;
8 num = [c/I2 k/I2];
9 den = [I1 I1*c*gamma/I2 I1*k*gamma/I2 0 0];
10 [aP, bP, cP, dP] = ND2SS( num, den );
11 wLead = 2
12 [aL, bL, cL, dL] = LeadLag( wLead, 80, 0 );
13 [a,b,c,d] = Series(aL,bL,cL,dL,aP,bP,cP,dP);
14 w = logspace(-2,1,1000);
15 [m,p] = FResp(a,b,c,d,1,w);
16 yL = {'Magnitude(db)' 'Phase(deg)'};
17 j = find(p >= 0);
18 p(j) = p(j) - 360;
19 Plot2D(w,[20*log10(m);p], 'Frequency(rad/s)',yL,'
    NonCollocated with Lead');

```



Example G.22: Bode plot with a flex compensator zero at 0.6 rad/s.

G.12. Model-following control

Model following is an approach in which the system is driven to follow a desired model. This approach is demonstrated using a single integrator as an example. The plant is

$$G(s) = \frac{1}{s} \quad (\text{G.180})$$

It is desired that the plant appears as

$$M(s) = \frac{1}{\tau s + 1} \quad (\text{G.181})$$

The following block diagram in Fig. G.10 shows a model-following controller. u is the command and is fed both to the plant and the model. The closed-loop transfer function is

$$\frac{y}{u} = \frac{\tau s + 1 + K}{(\tau s + 1)(s + K)} \quad (\text{G.182})$$

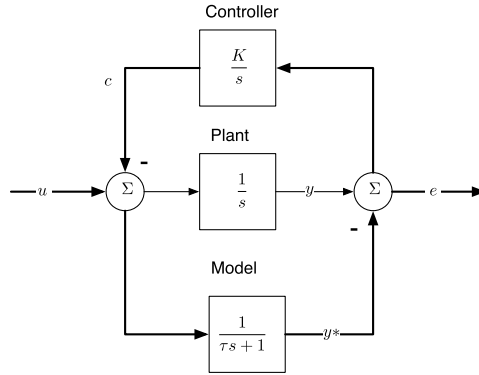


Figure G.10 Model-following controller.

We see that as $K \rightarrow \infty$, the transfer function approaches:

$$\frac{y}{u} = \frac{1}{\tau s + 1} \quad (\text{G.183})$$

For finite values of K the damping ratio and undamped natural frequency are

$$\omega_c = \sqrt{\frac{K}{\tau}} \quad (\text{G.184})$$

$$\zeta = \frac{K\tau + 1}{2\sqrt{K\tau}} \quad (\text{G.185})$$

It follows that the system is overdamped ($\zeta \geq 1$) when $K \geq 1/\tau$. The same response could be obtained using the simple feedback controller shown in Fig. G.11. The result-

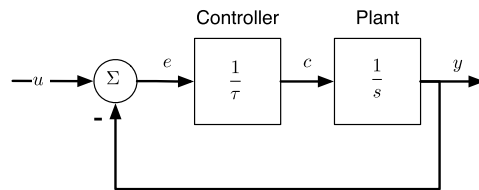


Figure G.11 Feedback controller.

ing transfer function is

$$\frac{y}{u} = \frac{1}{\tau s + 1} \quad (\text{G.186})$$

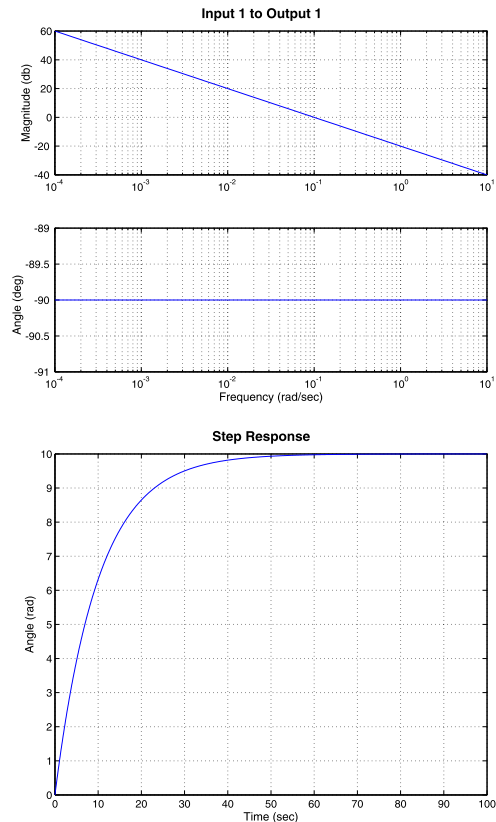
In this case, the desired response is achieved exactly with simpler constant-gain feedback. Therefore you would not use a model-following control in such a simple system. It is most useful in more complex situations and has been employed successfully on operational systems.

The loop-gain Bode plot and step response of the feedback controlled system are shown in Example G.23. The phase is constant at -90 deg, and the gain has a constant slope of -20 db/decade. This gives an infinite gain margin, since the phase never crosses 180 deg, and a phase margin of 90 deg.

```

1 % Plant
2 numP = 1;
3 denP = [1 0];
4 [aP,bP,cP,dP] = ND2SS( numP, denP );
5
6 % Controller
7 tau = 10;
8 K = 1/tau;
9
10 % Forward Loop Gain
11 [a,b,c,d] = Series( 0,0,0,K,aP,bP,cP,dP );
12
13 FResp(a,b,c,d);
14
15 % Closed Loop System
16 [aC,bC,cC,dC] = CLoopS( aP,bP,cP, K );
17
18 g = statespace( aC,bC,cC,dC );
19 dT = 0.1;
20 nSim = 1000;
21 y = Step( g, 1, dT, nSim );
22 t = (0:(nSim-1))*dT;
23 Plot2D( t, y, 'Time(sec)', 'Angle(rad)', 'Step
    Response' );

```



Example G.23: Model-following control.

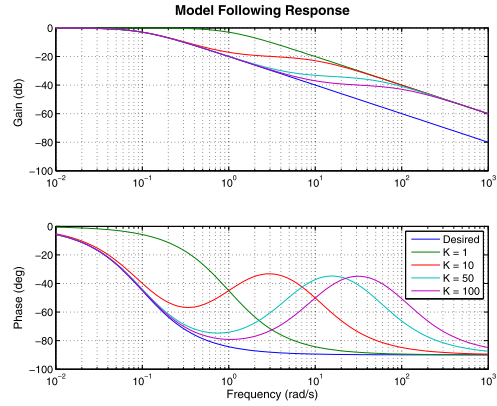
The Bode plot in Example G.24 shows the effect of the gain K on the control response. Here, we compare the desired response to the response of the closed-loop system associated with the model-following control of Fig. G.10. The closed-loop transfer function of the controlled system is given in (G.182). The controlled system matches the desired response at low frequencies, but a shift in magnitude and phase occurs at

high frequencies. For high gain, the model-matching controller approaches the model, pushing the location of this magnitude and phase shift to a higher frequency.

```

1 tau = 10;
2 K = [1 10 50 100];
3 w = logspace(-2,3,100);
4 m = zeros(length(K)+1,100);
5 p = zeros(length(K)+1,100);
6 [a,b,c,d] = ND2SS(1,[tau 1]); % desired
   response
7 [m(1,:),p(1,:)] = FResp(a,b,c,d,1,1,w);
8 for i=1:length(K)
9     num = [tau K(i)];
10    den = [tau tau*K(i)+1 K(i)];
11    [a,b,c,d] = ND2SS(num,den);
12    [m(i+1,:),p(i+1,:)] = FResp(a,b,c,d,1,1,w);
13 end
14
15
16
17 yL = {'Gain_(db)', 'Phase_(deg)'};
18 Plot2D( w, [20*log10(m);p], 'Frequency_(rad/s)',
        ....
        yL, 'ModelFollowingResponse', 'xlog
        ', {'[1:5]' '[6:10]'}))
19
20 legend('Desired', 'K_1', 'K_10', 'K_50', 'K_100')

```



Example G.24: Model-following control response with increasing gain.

G.13. Double-integrator control

G.13.1 Introduction

The simplest model for many vehicles is a double integrator. This is the equation

$$m\ddot{x} = F \quad (\text{G.187})$$

for linear motion where x equals position, m equals mass, and F equals force,

$$I\ddot{\theta} = T \quad (\text{G.188})$$

for rotational motion where θ equals angle, I inertia, and T equals torque. We can write both equations as

$$\ddot{w} = u \quad (\text{G.189})$$

where u is acceleration and w is the state. A step acceleration produces the response

$$w = \frac{1}{2}ut^2 + \dot{w}(0)t + w(0) \quad (\text{G.190})$$

which shows that w grows linearly with the initial time derivative and with the square of time multiplied by the acceleration. If u is a control this response is fine. If u is a disturbance we do not want w to change and so must apply control. Interestingly,

(G.190) is the basis for bang-bang control. That is you command u with one sign then when w reaches the half-way point, switch the sign of w . This will be discussed later.

In this section we will first discuss linear control techniques and then discuss the mathematical basis for bang-bang control. The system is illustrated in Fig. G.12.

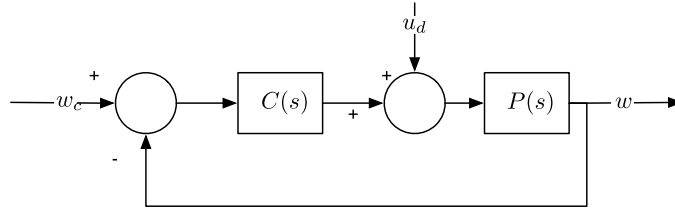


Figure G.12 Block diagram of the closed-loop control system.

The response of w to disturbances and control inputs is

$$w = Pu_d + PC(w_c - w) \quad (\text{G.191})$$

or

$$\frac{w}{u_d} = \frac{P}{1 + PC} \quad (\text{G.192})$$

$$\frac{w}{w_c} = \frac{PC}{1 + PC} \quad (\text{G.193})$$

In both cases, the denominator is $1 + PC$ but the numerators for the command input, w_c , and the disturbance input, u_d are different. For the former, if PC is much greater than 1 the output exactly follows the input. This can be done by letting $C = K/P$ and making K large. This is known as inverting the plant but requires ∞ energy. PC is the open-loop response, that is the series connection of controller and plant.

G.13.2 Linear control

The first thing to try is proportional control

$$\ddot{w} + \sigma^2 w = u_d \quad (\text{G.194})$$

where ω will be the oscillation frequency of the system. It is instrumental to look at the double-integrator and oscillator-frequency responses. The two transfer functions are

$$\frac{w(s)}{u(s)} = \frac{1}{s^2} \quad (\text{G.195})$$

$$\frac{w(s)}{u(s)} = \frac{1}{s^2 + \sigma^2} \quad (\text{G.196})$$

The first has a singularity at $s = j_0$. The second has a singularity at $s = j_\sigma$. At high frequencies, they have the same response $\frac{1}{s^2}$. At j_0 the second has the response $\frac{1}{\sigma^2}$, thus the wider the bandwidth of the controller (determined by σ) the smaller the response.

The infinite responses in these two cases mean that the system will absorb all of the input at those frequencies and as $t \rightarrow \infty$ the response will go to ∞ . From a practical point of view, it means the system will oscillate at σ , which is usually undesirable.

Since the oscillatory response is undesirable in most cases the next step is to add damping. This is done by adding a control term proportional to the derivative of w

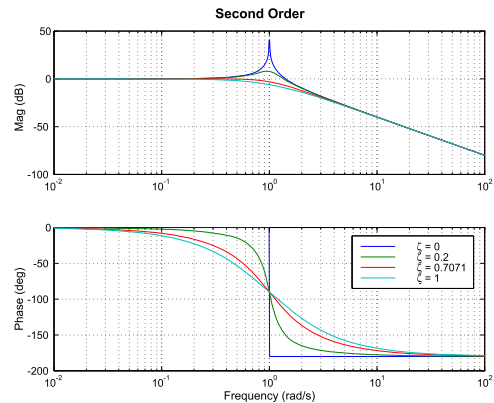
$$\ddot{w} + 2\zeta\sigma w + \sigma^2 w = u_d \quad (\text{G.197})$$

where ζ is the damping ratio that goes from 0 to ∞ . The transfer function is now

$$\frac{w(s)}{u(s)} = \frac{1}{s^2 + 2\zeta\sigma s + \sigma^2} \quad (\text{G.198})$$

```

1 sigma = 1;
2 zeta = [0 0.2 0.7071 1];
3 w = logspace(-2,2,1000);
4 for k = 1:length(zeta)
5     [aP, bP, cP, dP] = ND2SS( 1, [1 2*zeta(k)*sigma
6         sigma^2] );
7     [m(k,:), p(k,:)] = FResp(aP,bP,cP,dP,1,1,w);
8 end
9 yL = {'Mag_ (dB)' 'Phase_ (deg)'};
10 j = find(p > 0);
11 p(j) = p(j) - 360;
12 Plot2D(w,[20*log10(m);p], 'Frequency_ (rad/s)', yL, '
    Second_Order', 'xlog', {'[1:4]' '[5:8]'});
13 legend('\zeta=0', '\zeta=0.2', '\zeta=0.7071', '\zeta=1')
```



Example G.25: Second-order response.

When $\zeta = 1$ the resonant peak is gone. The trade for higher ζ is that there is more phase shift at lower frequencies. This means the control will be more sluggish. Note that at high frequencies the response is the same as for a double integrator, that is -180° and the magnitude decreases as $1/\omega^2$. This just means that the control has no effect at frequencies much higher than σ .

Adding damping has not changed the response to a step disturbance, which will still be $\frac{1}{\sigma^2}$. If this offset is unacceptable we must add an integrator to the controller, which will result in what is known as PID (proportional, integral, differential) control. There

are many ways to formulate the control but one is

$$\frac{u_c(s)}{w(s)} = K_p \left(1 + \tau_d s + \frac{1}{\tau_i s} \right) \quad (\text{G.199})$$

where τ_d is the rate time constant that is how long the system will take to damp and τ_i is how fast the system will integrate out a steady disturbance.

The open-loop transfer function is

$$\frac{w(s)}{u_d(s)} = \frac{K_p}{s^2} \left(1 + \tau_d s + \frac{1}{\tau_i s} \right) \quad (\text{G.200})$$

The closed-loop transfer function is

$$\frac{w(s)}{u_d(s)} = \frac{s}{s^3 + K_p \tau_d s^2 + K_p s + K_p / \tau_i} \quad (\text{G.201})$$

The desired closed-loop transfer function is

$$\frac{w(s)}{u_d(s)} = \frac{s}{(s + \gamma)(s^2 + 2\zeta \sigma s + \sigma^2)} \quad (\text{G.202})$$

or

$$\frac{w(s)}{u_d(s)} = \frac{s}{s^3 + (\gamma + 2\zeta \sigma)s^2 + \sigma(\sigma + 2\zeta \gamma)s + \gamma \sigma^2} \quad (\text{G.203})$$

The parameters are

$$K_p = \sigma(\sigma + 2\zeta \gamma) \quad (\text{G.204})$$

$$\tau_i = \frac{\sigma + 2\zeta \gamma}{\gamma \sigma} \quad (\text{G.205})$$

$$\tau_d = \frac{\gamma + 2\zeta \sigma}{\sigma(\sigma + 2\zeta \gamma)} \quad (\text{G.206})$$

It is interesting to evaluate the effect of the integrator. This is done in Example [G.26](#). The smaller γ , which is the inverse of the integrator time constant, the longer it takes to integrate out the disturbance.

The final form of the PID is shown below,

$$T = K_p + K_R \frac{\omega_R s}{s + \omega_R} + \frac{K_I}{s} \quad (\text{G.207})$$

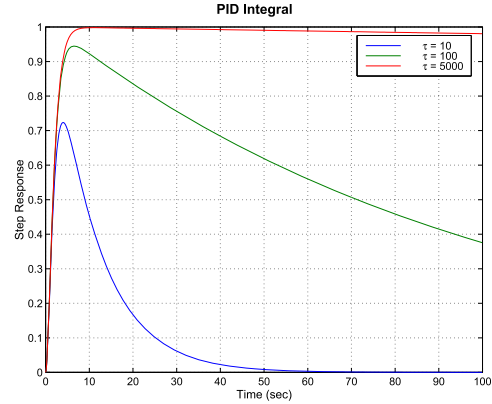
The derivative term is multiplied by a first-order filter to prevent differentiation at high frequencies. Rearranging gives

$$\frac{T}{\theta} = \frac{(K_p + K_R \omega_R)s^2 + (K_p \omega_R + K_I)s + (K_p + K_i \omega_R)}{s(s + \omega_R)} \quad (\text{G.208})$$

```

1 gamma = 1./[10 100 1000];
2 dT = 0.1;
3 nSim = 1000;
4 y = zeros(3,nSim);
5 for k = 1:length(gamma)
6   [a,b,c,d] = ND2SS( [1 0], [1 (gamma(k) + 2) (1
   + 2*gamma(k)) gamma(k)]);
7   g = statespace(a,b,c,d);
8   y(k,:) = Step( g, 1, dT, nSim );
9 end
10 t = (0:(nSim-1))*dT;
11 Plot2D(t, y, 'Time(sec)', 'StepResponse', 'PID
   Integral');
12 legend('\gamma=0.1', '\gamma=0.01', '\gamma=0.001')

```



Example G.26: Effect of the integrator on a second-order step response.

If ω_R is set to ∞ this becomes

$$\frac{T}{\theta} = \frac{K_R s^2 + K_P s + K_I}{s} \quad (\text{G.209})$$

This transfer function is not proper, i.e., it goes to ∞ as s goes to 0. The latter cannot be implemented in state-space form for this reason. In practice, this is not a problem if noise filtering is included in the loop. The built-in rate filter is often sufficient. On occasion, it may be necessary to add a notch filter to notch out a flex or slosh mode. A low-pass filter may also be added if the sensor is noisy.

G.13.3 Phase-plane controller

Another way to control a double integrator is a phase-plane controller. This is just the embodiment of the equation

$$x = \frac{1}{2} u t^2 + v(0)t + x(0) \quad (\text{G.210})$$

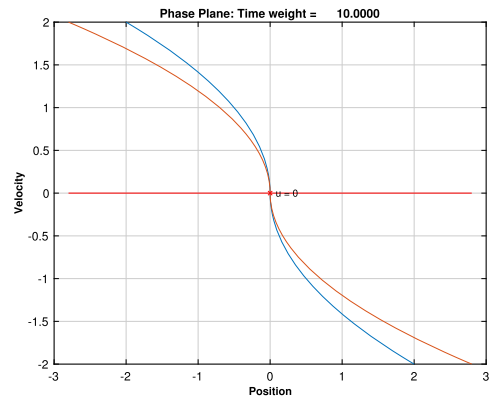
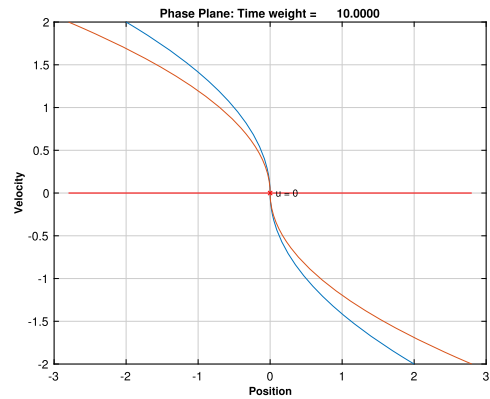
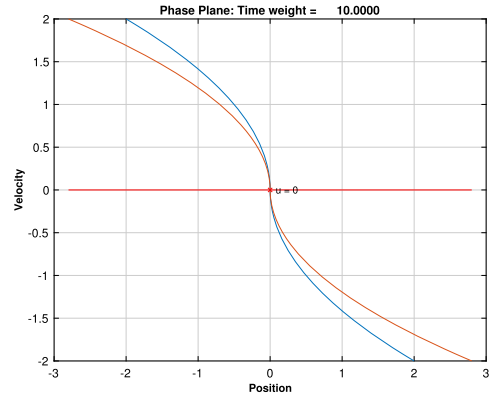
where $\dot{x} = v$ when u is limited to be $\pm u_{\max}$. There will be a locus of points $[x, v]$ for which a constant u will drive x and v to zero. This is a switching curve. If we are only solving the time-optimal problem there will be one switching curve. If we solve the fuel-time optimal problem, there will be two switching curves and the trajectory will coast between the curves.

Example G.27 shows a fuel-time optimal pair of switching curves and then compares a PID controller with a phase-plane controller.

```

1 PhasePlane( [], [], 10);
2 dT = 0.01;
3 [a,b,c,d] = PIDMIMO( 1, 0.7071, 0.1, 100.0,
4     5.0, dT, 'Z' );
5 aP = [0 1; 0 0]; bP = [0;1]; cP = [1
6     0]; dP = 0;
7 [aP,bP] = C2DZOH( aP, bP, dT );
8 xPID = [1;0];
9 nSim = 8000;
10 xPID = [xPID zeros(2,nSim)];
11 xPP = xPID;
12 xC = zeros(2,1);
13 uPID = zeros(1,nSim+1);
14 uPP = zeros(1,nSim+1);
15
16 for k = 1:nSim
17     % PID Controller
18     u = -c*xC - d*xPID(1,k);
19     xC = a*xC + b*xPID(1,k);
20     xPID(:,k+1) = aP*xPID(:,k) + bP*u;
21     uPID(:,k+1) = u;
22
23     % Phase Plane
24     u = PhasePlane(xPP(2,k),xPP(1,k),0,0.01,0.01);
25     xPP(:,k+1) = aP*xPP(:,k) + bP*u;
26     uPP(:,k+1) = u;
27
28 t = (0:nSim)*dT;
29 Plot2D( [xPID(1,:);xPP(1,:)], [xPID(2,:);xPP(2,:)]
30     , 'x', 'v', 'PID' );
31 legend('PID', 'PhasePlane')
32 Plot2D( t, [xPID;xPP;uPID;uPP], 'Time(sec)', {'x'
33     'v' 'uPID' 'uPP'}....
34     'Red is the PhasePlane', 'lin', {'[1 3]'
35     '[2 4]' '[5 6]'} );

```



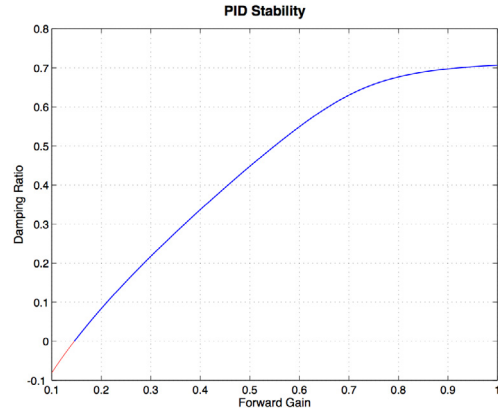
Example G.27: Phase-plane controller.

G.13.4 Control limiting

A PID controller is only conditionally stable. If the forward gain decreases due to actuator saturation, the controller can be unstable. This is sometimes known as integrator windup. This is seen in Example G.28. For a forward gain of less than 0.14 the system is unstable. The unstable region is shown in red.

```

1 inr = 10000;
2
3 [aC, bC, cC, dC] = PIDMIMO( inr, 0.7071, 0.04,
4                             50, 1.0 );
5 aP = [0 1; 0 0];
6 bP = [0; 1/inr];
7 cP = [1 0];
8 dP = 0;
9 [a, b, c, d] = Series(aP, bP, cP, dP, aC, bC, cC, dC);
10 kF = linspace(0.1, 1);
11
12 for k = 1:length(kF)
13     s = eig(a - kF(k)*b*c);
14     z = S2Damp(s);
15     zeta(k) = z(3);
16 end
17 j = find(zeta <= 0);
18 k = j(end):length(zeta);
19 Plot2D(kF(k), zeta(k), 'Forward Gain', 'Damping
20 Ratio', 'PID Stability')
21 hold on
22 plot(kF(j), zeta(j), 'r')
```



Example G.28: PID damping ratio.

Since a PD controller is unconditionally stable the simplest way to deal with actuator saturation is to limit the integral. This can be done for any state-space system. If u is the actuator output write

$$y_k = \text{sat}(cx_k + du_k) \quad (\text{G.211})$$

$$x_{k+1} = (a - lc)x_k + (b - ld)u_k + ly_k \quad (\text{G.212})$$

If y is not saturated these equations reduce to

$$y_k = x_k + du_k \quad (\text{G.213})$$

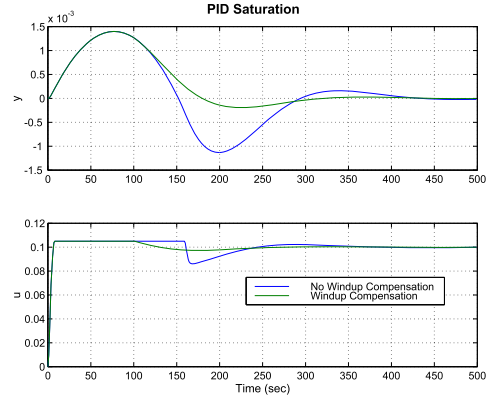
$$x_{k+1} = ax_k + bu_k \quad (\text{G.214})$$

The gain matrix l is chosen so that the eigenvalues of $a - lc$ are all zero, that is it is a deadbeat controller, see Example G.29. Note that the performance of the compensator with windup compensation is superior to that of the compensator without the windup compensation. The uncompensated system effectively looks more lightly damped than that with the windup compensation.

```

1 dT = 1.0; inr = 10000;
2
3 [aC, bC, cP, dP] = PIDMIMO( inr(1,1), ...
4                             0.7071, ...
5                             0.04, 50, ...
6                             1.0, dT, 'z' );
7 [aP, bP] = C2DZOH( [0 1; 0 0], [0; 1/inr(1,1)], dT
8 );
9 nSim = 500;
10 uStep = 0.1; uSat = 0.105;
11 uPlot = zeros(2, nSim);
12 yPlot = zeros(2, nSim);
13
14 for j = 1:2
15     x = zeros(2,1);
16     y = 0; xPID = [0;0];
17
18     if ( j == 2 )
19         [aC, bC, lC] = Windup( aC, bC, cC, dC );
20     else
21         lC = [0;0];
22     end
23
24     for k = 1:nSim
25         % Controller
26         %-----
27         u = cC*xPID + dC*y;
28         if ( abs(u) > uSat ) u = sign(u)*uSat; end
29
30         xPID = aC*xPID + bC*y + lC*u;
31
32         yPlot(j,k) = y; uPlot(j,k) = u;
33
34         % The plant
35         %-----
36         x = aP*x + bP*(uStep-u);
37         y = x(1);
38     end
39 end
40
41 Plot2D( (0:(nSim-1))*dT, [ yPlot; uPlot ], 'Time (sec)'
42         , [ 'y'; 'u' ], ...
43         , 'PID Saturation', 'lin', [ '1:2'; '3:4' ] );
44 legend( 'No Windup Compensation', 'Windup
45         Compensation' )
46 PrintFig(1,1,1, 'PIDWindup')

```



Example G.29: Windup compensation.

G.13.5 Cross-axis coupling

It is often the case that multiple axes must be controlled. Given a system of the form

$$M\ddot{x} = F \quad (\text{G.215})$$

where M is not diagonal we define F as

$$F = Ma \quad (\text{G.216})$$

where a is the vector of accelerations. Then, our control problem is the decoupled problem,

$$\ddot{x} = a \quad (\text{G.217})$$

so we can solve three decoupled PID problems.

G.14. Lyapunov control

G.14.1 Background

Systems that are time invariant are known as autonomous systems. Those that are time varying are called nonautonomous systems,

$$\dot{x} = f(x) \quad (\text{G.218})$$

is an autonomous system

$$\dot{x} = f(x, t) \quad (\text{G.219})$$

is nonautonomous. Adding a controller

$$\dot{x} = f(x, u) \quad (\text{G.220})$$

can make an autonomous system nonautonomous if $u(t)$. For example, a tracking controller is nonautonomous since the set point varies with time. Systems with external disturbances are nonautonomous.

G.14.2 Theory

Lyapunov's control theory consists of two methods: Lyapunov's first method that involves linearization about an operating point and Lyapunov's direct method that applies to nonlinear systems. Lyapunov's direct method is based on the idea that if a system dissipates energy it will eventually reach equilibrium.

Lyapunov's direct method has two steps. The first is to find a scalar function of the states of the system, known as the Lyapunov function, and then to evaluate its derivative along the trajectory of the system. If the Lyapunov function is always decreasing the system will eventually reach equilibrium.

If there exists a Lyapunov function $V(x)$ such that

1. $V(x)$ is positive-definite;
2. $\dot{V}(x)$ is negative-definite;
3. $V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$;

then the equilibrium point is uniformly asymptotically stable.

For a linear system

$$\dot{x} = Ax + Bu \quad (\text{G.221})$$

if we define the cost

$$C = \frac{1}{2} (x^T Q x + u^T R u) \quad (\text{G.222})$$

the solution, P , to the matrix Riccati equation

$$A^T P + PA - PBR^{-1}B^T P - Q = 0 \quad (\text{G.223})$$

can be used in the Lyapunov function

$$V = x^T P x \quad (\text{G.224})$$

and the stabilizing controller is

$$u = -R^{-1} B^T P x \quad (\text{G.225})$$

G.14.3 Nonlinear rate damper

The dynamical equations for a spacecraft are

$$I\dot{\omega} + \omega^\times I\omega = T \quad (\text{G.226})$$

Let the control be

$$T_c = -K\omega \quad (\text{G.227})$$

The dynamical equations are now

$$I\dot{\omega} + \omega^\times I\omega + K\omega = T_d \quad (\text{G.228})$$

where T_d is the disturbance torque. The kinetic energy is our Lyapunov function and is

$$V = \frac{1}{2} \omega^T I \omega \quad (\text{G.229})$$

The time derivative is

$$\dot{V} = \omega^T I \dot{\omega} \quad (\text{G.230})$$

Substitute in the dynamical equations

$$\dot{V} = \omega^T (T_d - (\omega^\times I + K)\omega) \quad (\text{G.231})$$

Looking at just the damping case in which $T_d = 0$,

$$\dot{V} = -\omega^T (\omega^\times I + K) \omega \quad (\text{G.232})$$

This system is autonomous if I is time invariant. $\dot{V}$ is negative-definite if $-\dot{V}$ is positive-definite, which requires that for all $x \neq 0$ $\dot{V} > 0$. If we linearize this we see that any positive diagonal gain matrix makes this true.

$$K_x \omega_x^2 + K_y \omega_y^2 + K_z \omega_z^2 > 0 \quad (\text{G.233})$$

for all ω .

We can expand this for a diagonal inertia matrix and diagonal gain matrix without a loss of generality. Expanding, we find

$$\omega^T (\omega^\times I + K) \omega = \omega_x \omega_y \omega_z [I_z - I_y + I_x - I_z + I_y - I_x] + K_x \omega_x^2 + K_y \omega_y^2 + K_z \omega_z^2 \quad (\text{G.234})$$

which equals

$$K_x \omega_x^2 + K_y \omega_y^2 + K_z \omega_z^2 \quad (\text{G.235})$$

G.15. First- and second-order systems

A damped first-order system is

$$\dot{x} + ax = au \quad (\text{G.236})$$

where a is the damping constant, x is the state, and u is the input. When the initial transient is done, $\dot{x} = 0$, the steady-state response is

$$x = u \quad (\text{G.237})$$

A damped second-order system is

$$\ddot{x} + 2\zeta\omega\dot{x} + \omega^2x = \omega^2u \quad (\text{G.238})$$

where ζ is the damping ratio, x is the state, ω is the undamped natural frequency, and u is the input. When the initial transient is done, $\dot{x} = 0$ and $\ddot{x} = 0$, the steady-state response is

$$x = u \quad (\text{G.239})$$

G.16. Inner and outer loops

Assume you have a dynamical system composed of two equations for the pitch loop of a momentum-bias spacecraft.

$$T = I\ddot{\theta} + J\dot{\Omega} \quad (\text{G.240})$$

$$T_w = J\dot{\Omega} \quad (\text{G.241})$$

where I is the pitch inertia, J is the momentum wheel inertia, T is an external disturbance torque, θ is the angle that you want to control, Ω is your momentum wheel speed, and T_w is the torque on the wheel produced by an electric motor connecting the

wheel to the spacecraft. You have a measurement of Ω and θ but not T_w . How do we connect the two? Write

$$J\dot{\Omega} = I(2\zeta\sigma\dot{\theta} + \sigma^2\theta) \quad (\text{G.242})$$

where ζ is the damping ratio and σ is the undamped natural frequency. Our first dynamical equation is now

$$T = I\ddot{\theta} + I(2\zeta\sigma\dot{\theta} + \sigma^2\theta) \quad (\text{G.243})$$

which is a damped second-order system. Divide by I the clarify to equation

$$\frac{T}{I} = \ddot{\theta} + \zeta\sigma\dot{\theta} + \sigma^2\theta \quad (\text{G.244})$$

Our inner-loop commands wheel speed,

$$T_w = Jk(\Omega_c - \Omega) \quad (\text{G.245})$$

where k is the gain and Ω_c is the commanded wheel speed. The inner-loop dynamical equation is now

$$k\Omega_c = \dot{\Omega} + k\Omega \quad (\text{G.246})$$

which is a damped first-order system. If k is high enough, that is the inner loop is very fast, the wheel speed will equal the commanded speed.

$$\Omega = \Omega_c \quad (\text{G.247})$$

This is the “steady-state” response. Ω is determined by integrating Eq. (G.242)

$$\Omega = \left(\frac{I}{J}\right) \int (2\zeta\sigma\dot{\theta} + \sigma^2\theta) \quad (\text{G.248})$$

or

$$\Omega_c = \left(\frac{I}{J}\right) \sigma \left(2\zeta\dot{\theta} + \sigma \int \theta\right) \quad (\text{G.249})$$

Our outer loop will control θ . Our inner loop will control Ω , which is > 0 for all ω . Therefore the damping system will work for the nonlinear system.